

CAIRO SECURITY CLAN

ATOMIQ LABS EVM UPDATE

SECURITY ASSESSMENT REPORT

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Prepared for

ATOMIQ LABS

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1 About Cairo Security Clan

Cairo Security Clan is a leading force in the realm of blockchain security, dedicated to fortifying the foundations of the digital age. As pioneers in the field, we specialize in conducting meticulous smart contract security audits, ensuring the integrity and reliability of decentralized applications built on blockchain technology.

At Cairo Security Clan, we boast a multidisciplinary team of seasoned professionals proficient in blockchain security, cryptography, and software engineering. With a firm commitment to excellence, our experts delve into every aspect of the Web3 ecosystem, from foundational layer protocols to application-layer development. Our comprehensive suite of services encompasses smart contract audits, formal verification, and real-time monitoring, offering unparalleled protection against potential vulnerabilities.

Our team comprises industry veterans and scholars with extensive academic backgrounds and practical experience. Armed with advanced methodologies and cutting-edge tools, we scrutinize and analyze complex smart contracts with precision and rigor. Our track record speaks volumes, with a plethora of published research papers and citations, demonstrating our unwavering dedication to advancing the field of blockchain security.

At Cairo Security Clan, we prioritize collaboration and transparency, fostering meaningful partnerships with our clients. We believe in a customer-oriented approach, engaging stakeholders at every stage of the auditing process. By maintaining open lines of communication and soliciting client feedback, we ensure that our solutions are tailored to meet the unique needs and objectives of each project.

Beyond our core services, Cairo Security Clan is committed to driving innovation and shaping the future of blockchain technology. As active contributors to the ecosystem, we participate in the development of emerging technologies such as Starknet, leveraging our expertise to build robust infrastructure and tools. Through strategic guidance and support, we empower our partners to navigate the complexities of the blockchain landscape with confidence and clarity.

In summary, Cairo Security Clan stands at the forefront of blockchain security, blending technical prowess with a client-centric ethos to deliver unparalleled protection and peace of mind in an ever-evolving digital landscape. Join us in safeguarding the future of decentralized finance and digital assets with confidence and conviction.

2 Disclaimer

Disclaimer Limitations of this Audit:

This report is based solely on the materials and documentation provided by you to Cairo Security Clan for the specific purpose of conducting the security review outlined in the Summary of Audit and Scoped Files. The findings presented here may not be exhaustive and may not identify all potential vulnerabilities. Cairo Security Clan provides this review and report on an ”as-is” and ”as-available” basis. You acknowledge that your use of this report, including any associated services, products, protocols, platforms, content, and materials, occurs entirely at your own risk.

Inherent Risks of Blockchain Technology:

Blockchain technology remains in its developmental stage and is inherently susceptible to unknown risks and vulnerabilities. This review is specifically focused on the smart contract code and does not extend to the compiler layer, programming language elements beyond the reviewed code, or other potential security risks outside the code itself.

Report Purpose and Reliance:

This report should not be construed as an endorsement of any specific project or team, nor does it guarantee the absolute security of the audited smart contracts. No third party should rely on this report for any purpose, including making investment or purchasing decisions.

Liability Disclaimer:

To the fullest extent permitted by law, Cairo Security Clan disclaims all liability associated with this report, its contents, and any related services and products arising from your use. This includes, but is not limited to, implied warranties of merchantability, fitness for a particular purpose, and non-infringement.

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FOR THE AVOIDANCE OF DOUBT, THIS REPORT, ITS CONTENT, ACCESS, AND/OR USE, INCLUDING ANY ASSOCIATED SERVICES OR MATERIALS, SHOULD NOT BE CONSIDERED OR RELIED UPON AS FINANCIAL, INVESTMENT, TAX, LEGAL, REGULATORY, OR OTHER PROFESSIONAL ADVICE.

3 Executive Summary

This document presents the security review performed by Cairo Security Clan on the Atomiq Exchange.

On this review, the assessment was conducted on the specific commits or pull requests provided. The full codebase was not reviewed.

atomiq.exchange is a fully trustless cross-chain decentralized exchange (DEX) allowing you to swap between smart chains and Bitcoin, without having to trust any intermediary in the process.

All transactions are processed atomically with strong security guarantees based on bitcoin light client (leveraging bitcoin's proof-of-work security) & submarine swaps (leveraging HTLCs - hash-time locked contracts over bitcoin's lightning network). With this approach atomiq.exchange is able to offer security guarantees far exceeding those of existing bridging or cross-chain swapping solutions. Learn more from docs and FAQ.

The audit was performed using

- manual analysis of the codebase,
- automated analysis tools,
- simulation of the smart contract,
- analysis of edge test cases

There are 2 points of attention, where 0 are classified as Critical, 1 as High, 1 as Medium, 0 as Low, 0 as Informational, 0 as Best Practices, and 0 as Undetermined. The issues are summarized in Fig. 1.

This document is organized as follows. Section 1 About Cairo Security Clan. Section 2 Disclaimer. Section 3 Executive Summary. Section 4 Summary of Audit. Section 5 Risk Classification. Section 6 Issues by Severity Levels. Section 7 Test Evaluation.

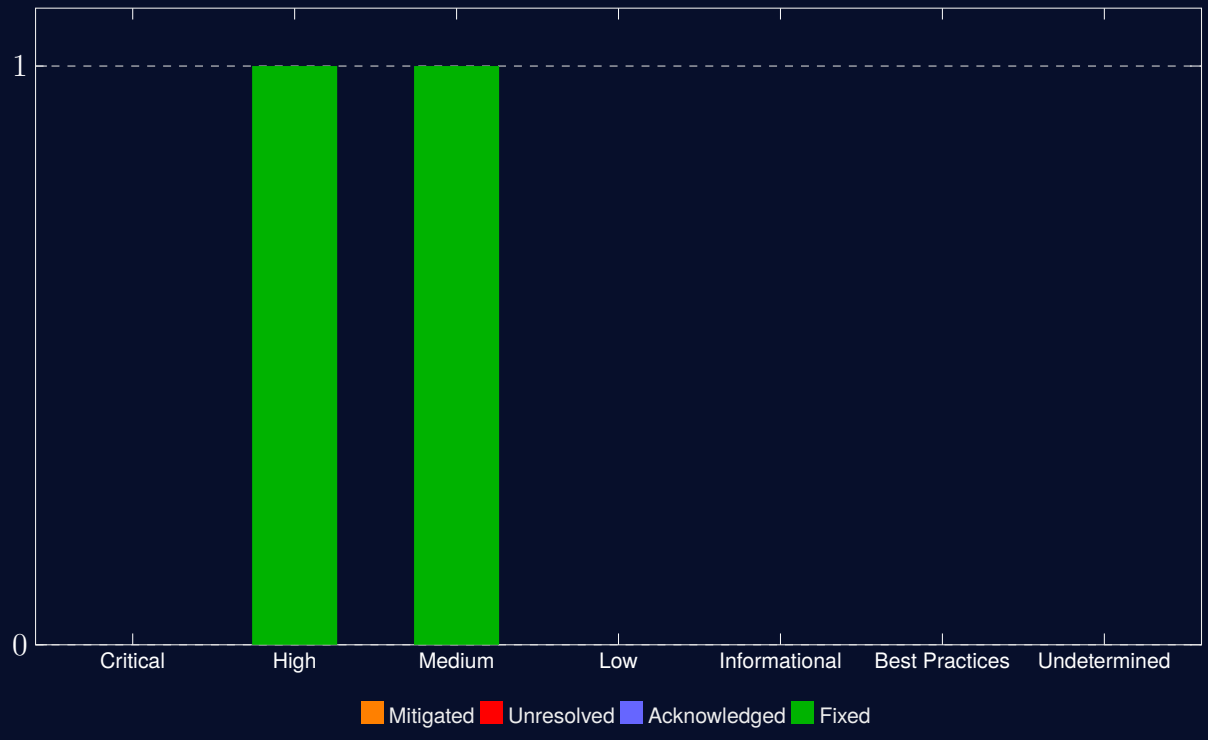


Fig 1: Distribution of issues: Critical (0), **High** (1), **Medium** (1), **Low** (0), **Informational** (0), **Best Practices** (0), **Undetermined** (0). **Distribution of status: Fixed** (2), **Acknowledged** (0), **Mitigated** (0), **Unresolved** (0).

4 Summary of Audit

Audit Type	Security Review Between Commits (Best Effort)
Solidity Version	0.8.28
Final Report	17/12/2025
Repository	atomiqlabs/atomiq-contracts-evm
PR #3	PR #3
PR #4	PR #4
PR #5	PR #5
Documentation	Website documentation
Test Suite Assessment	High

4.1 Scoped Files

	Contracts
1	contracts/escrow_manager/EscrowManager.sol
2	contracts/escrow_manager/components/LpVault.sol
3	contracts/execution_contract/ExecutionContract.sol
4	contracts/execution_proxy/ExecutionProxy.sol
5	contracts/execution_proxy/Executor.sol
6	contracts/spv_swap_vault/SpvVaultManager.sol
7	contracts/transfer_utils/TransferHandler.sol
8	contracts/transfer_utils/interfaces/IDepositOnlyWETH.sol
9	contracts/utils/ContractCallUtils.sol

4.2 Issues

	Findings	Severity	Update
1	`trySafeTransfer` ignores `token` parameter and always transfers from WETH contract	High	Fixed
2	Unreachable fallback logic in `trySafeTransfer` causes non-standard ERC20 tokens to always fail	Medium	Fixed

5 Risk Classification

The risk rating methodology used by Cairo Security Clan follows the principles established by the CVSS risk rating methodology. The severity of each finding is determined by two factors: **Likelihood** and **Impact**.

Likelihood measures how likely an attacker will uncover and exploit the finding. This factor will be one of the following values:

a) **High**: The issue is trivial to exploit and has no specific conditions that need to be met;

b) **Medium**: The issue is moderately complex and may have some conditions that need to be met;

c) **Low**: The issue is very complex and requires very specific conditions to be met.

When defining the likelihood of a finding, other factors are also considered. These can include but are not limited to Motive, opportunity, exploit accessibility, ease of discovery, and ease of exploit.

Impact is a measure of the damage that may be caused if an attacker exploits the finding. This factor will be one of the following values:

a) **High**: The issue can cause significant damage such as loss of funds or the protocol entering an unrecoverable state;

b) **Medium**: The issue can cause moderate damage such as impacts that only affect a small group of users or only a particular part of the protocol;

c) **Low**: The issue can cause little to no damage such as bugs that are easily recoverable or cause unexpected interactions that cause minor inconveniences.

When defining the impact of a finding other factors are also considered. These can include but are not limited to Data/state integrity, loss of availability, financial loss, and reputation damage. After defining the likelihood and impact of an issue, the severity can be determined according to the table below.

		Likelihood		
		High	**Medium**	**Low**
Impact	**High**	Critical	High	Medium
	Medium	High	Medium	Low
	Low	Medium	Low	Info/Best Practices

To address issues that do not fit a High/Medium/Low severity, Cairo Security Clan also uses three more finding severities: **Informational**, **Best Practices** and **Gas**

a) **Informational** findings do not pose any risk to the application, but they carry some information that the audit team intends to formally pass to the client;

b) **Best Practice** findings are used when some piece of code does not conform with smart contract development best practices;

c) **Gas** findings are used when some piece of code uses more gas than it should be or have some functions that can be removed to save gas.

6 Issues by Severity Levels

6.1 High

6.1.1 `trySafeTransfer` ignores `token` parameter and always transfers from WETH contract

File(s): contracts/transfer_utils/TransferHandler.sol

Description: The `trySafeTransfer()` function accepts a `token` parameter but never uses it. Instead, it hardcodes `_wrappedEthContract` as the target of the IERC20.transfer call:

```
function trySafeTransfer(address token, address dst, uint256 amount) private returns (bool success) {
    //Attempt a transfer using the safe call
    (bool wethTransferSuccess, bytes memory returnData) = ContractCallUtils.safeCall(
        address(_wrappedEthContract),
        0,
        abi.encodeWithSelector(IERC20.transfer.selector, dst, amount)
    );
```

This function is called from `_TokenHandler_transferOutNoRevert()` for both native-token fallback (line 117) and ERC20 transfers (line 119). When an ERC20 transfer is attempted, the code passes the correct token address but `trySafeTransfer` ignores it and calls `_wrappedEthContract.transfer(dst, amount)` instead. This causes: 1. The intended token is never transferred 2. WETH is transferred (or attempted) instead, potentially draining the contract's WETH balance

Recommendation(s): Update the `trySafeTransfer()` function to call the provided `token` address, not `_wrappedEthContract`. Add tests to cover ERC20 transfers for both native-wrap fallback and arbitrary tokens to prevent regressions.

Status: Fixed

Update from the client: Fixed in commit bcc3b5e

6.2 Medium

6.2.1 Unreachable fallback logic in `trySafeTransfer` causes non-standard ERC20 tokens to always fail

File(s): contracts/transfer_utils/TransferHandler.sol

Description: The assembly block in `trySafeTransfer` attempts to handle tokens that do not return data (a common non-standard behavior). However, the fallback branch contains a logical error that makes it unreachable: Tokens that return no data or short/absent booleans are always classified as failed transfers even when the call succeeded, making `_TokenHandler_transferOutNoRevert` unreliable.

```
assembly ("memory-safe") {
    let returnDataSize := mload(returnData)
    let returnValue := mload(add(returnData, 32))
    success := and(
        gt(returnDataSize, 31), //Return data size is at least 32 bytes
        eq(returnValue, 1) //Returned value is true (1)
    )
    //If call success all is good.
    //Otherwise (not success is not true), we need to perform further checks
    if iszero(success) {
        //If the return value is not true, then the call is only successful if:
        // - the token address has code
        // - the returndata is empty
        success := and(success, and(iszero(returndatasize()), gt(extcodesize(token), 0)))
    }
}
```

The if `iszero(success)` block only runs when `success` is already false. Inside that block, the code performs `success := and(success, ...)`. Since `success` is false and false AND anything equals false, the result is always false regardless of the second operand. The intended "no return data but contract has code" check can never flip success to true. As a result, any token that returns no data (e.g., USDT on some chains, or older non-compliant tokens) will be treated as a failed transfer even when the underlying call succeeded.

Recommendation(s): Rewrite the return-data validation to follow the OpenZeppelin pattern

Status: Fixed

Update from the client: Fixed in commit bcc3b5e

7 Test & Compilation Evaluation

7.1 Compilation Output

```
npx hardhat compile
Warning: Unused function parameter. Remove or comment out the variable name to silence this warning.
  --> contracts/test_contracts/BrokenTransferWETH9.sol:11:23:
   |
11 |     function transfer(address to, uint256 amount) public pure override returns (bool) {
   |                       ^^^^^^^^^^


Warning: Unused function parameter. Remove or comment out the variable name to silence this warning.
  --> contracts/test_contracts/BrokenTransferWETH9.sol:11:35:
   |
11 |     function transfer(address to, uint256 amount) public pure override returns (bool) {
   |                                   ^^^^^^^^^^^^^^


Generating typings for: 113 artifacts in dir: typechain-types for target: ethers-v6
Successfully generated 320 typings!
Compiled 102 Solidity files successfully (evm target: cancun).
```

7.2 Tests Output

```
npx hardhat test
  BtcRelay
    PASS Valid constructor (657ms)
    PASS Invalid block PoW (blockhash not lower than target) (46ms)
    PASS Invalid block nbits (not same as last block) (81ms)
    PASS Invalid block nbits at difficulty adjustment (invalidly computed)
    PASS Invalid block previous blockhash (49ms)
    PASS Invalid block median timestamps (74ms)
    PASS Invalid block future timestamp (61ms)
[submitMainAndAssert] Gas usage, num headers: 20 gas used: 572340 gas per header: 28617
    PASS Valid main chain (457ms)
    PASS Invalid main chain, stored blockheader not committed (77ms)
[submitMainAndAssert] Gas usage, num headers: 4 gas used: 139268 gas per header: 34817
    PASS Invalid main chain, stored header is not the tip (92ms)
[submitMainAndAssert] Gas usage, num headers: 20 gas used: 572304 gas per header: 28615
    PASS Invalid short fork, stored header not in main chain (478ms)
[submitMainAndAssert] Gas usage, num headers: 20 gas used: 572328 gas per header: 28616
    PASS Invalid long fork, stored header not in main chain (513ms)
[submitMainAndAssert] Gas usage, num headers: 20 gas used: 572280 gas per header: 28614
[submitLongForkAndAssert] Gas usage, num headers: 4 gas used: 183234 gas per header: 45809
    PASS Invalid long fork, stored header not in fork state (623ms)
[submitMainAndAssert] Gas usage, num headers: 20 gas used: 572352 gas per header: 28618
[submitLongForkAndAssert] Gas usage, num headers: 4 gas used: 183198 gas per header: 45800
    PASS Invalid long fork, stored header not fork state tip (664ms)
[submitMainAndAssert] Gas usage, num headers: 135 gas used: 3685023 gas per header: 27296
[submitShortForkAndAssert] Gas usage, num headers: 134 gas used: 1371389 gas per header: 10234
    PASS Invalid main chain, re-org & try to build on top of now future blockheight (18592ms)
[submitMainAndAssert] Gas usage, num headers: 135 gas used: 3685155 gas per header: 27297
[submitShortForkAndAssert] Gas usage, num headers: 134 gas used: 1371365 gas per header: 10234
    PASS Invalid short fork, re-org & try to fork again from now future blockheight (17553ms)
[submitMainAndAssert] Gas usage, num headers: 135 gas used: 3685095 gas per header: 27297
[submitShortForkAndAssert] Gas usage, num headers: 134 gas used: 1371233 gas per header: 10233
    PASS Invalid long fork, re-org & try to fork again from now future blockheight (18981ms)
[submitMainAndAssert] Gas usage, num headers: 20 gas used: 572340 gas per header: 28617
[submitShortForkAndAssert] Gas usage, num headers: 20 gas used: 301996 gas per header: 15100
    PASS Valid short fork (831ms)
[submitMainAndAssert] Gas usage, num headers: 20 gas used: 572340 gas per header: 28617
[submitLongForkAndAssert] Gas usage, num headers: 20 gas used: 644828 gas per header: 32241
    PASS Valid long fork (784ms)
[submitMainAndAssert] Gas usage, num headers: 20 gas used: 572340 gas per header: 28617
[submitLongForkAndAssert] Gas usage, num headers: 4 gas used: 183246 gas per header: 45812
[submitLongForkAndAssert] Gas usage, num headers: 16 gas used: 544618 gas per header: 34039
    PASS Valid long fork in 2 txns (783ms)
[submitMainAndAssert] Gas usage, num headers: 135 gas used: 3684999 gas per header: 27296
[submitShortForkAndAssert] Gas usage, num headers: 134 gas used: 1371137 gas per header: 10232
    PASS Valid short fork, higher chainwork but shorter chain (16504ms)
[submitMainAndAssert] Gas usage, num headers: 20 gas used: 572292 gas per header: 28615
    PASS Invalid short fork, not enough length (406ms)
[submitMainAndAssert] Gas usage, num headers: 20 gas used: 572328 gas per header: 28616
[submitLongForkAndAssert] Gas usage, num headers: 5 gas used: 210792 gas per header: 42158
    PASS Invalid long fork, not enough length (469ms)
[submitMainAndAssert] Gas usage, num headers: 123 gas used: 3360327 gas per header: 27320
    PASS Invalid short fork, not enough chainwork, but long enough (8565ms)
[submitMainAndAssert] Gas usage, num headers: 123 gas used: 3360327 gas per header: 27320
[submitLongForkAndAssert] Gas usage, num headers: 124 gas used: 3477136 gas per header: 28041
    PASS Invalid long fork, not enough chainwork, but long enough (13749ms)

  EscrowManager
    Claim
      PASS Valid claim (payOut=false,reputation=false,securityDeposit=false,claimerBounty=false,usesNativeToken=
     false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false) (110ms)
      PASS Valid claim (payOut=true,reputation=false,securityDeposit=false,claimerBounty=false,usesNativeToken=
     false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
      PASS Valid claim (payOut=false,reputation=true,securityDeposit=false,claimerBounty=false,usesNativeToken=
     false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
      PASS Valid claim (payOut=true,reputation=true,securityDeposit=false,claimerBounty=false,usesNativeToken=
     false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
```

```
64   PASS Valid claim (payOut=false,reputation=false,securityDeposit=true,claimerBounty=false,usesNativeToken=
     false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
65   PASS Valid claim (payOut=true,reputation=false,securityDeposit=true,claimerBounty=false,usesNativeToken=
     false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
66   PASS Valid claim (payOut=false,reputation=true,securityDeposit=true,claimerBounty=false,usesNativeToken=
     false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
67   PASS Valid claim (payOut=true,reputation=true,securityDeposit=true,claimerBounty=false,usesNativeToken=
     false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
68   PASS Valid claim (payOut=false,reputation=false,securityDeposit=false,claimerBounty=true,usesNativeToken=
     false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
69   PASS Valid claim (payOut=true,reputation=false,securityDeposit=false,claimerBounty=true,usesNativeToken=
     false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
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     false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
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     false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
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     false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
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     false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
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     false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
75   PASS Valid claim (payOut=true,reputation=true,securityDeposit=true,claimerBounty=true,usesNativeToken=false
     ,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
76   PASS Valid claim (payOut=false,reputation=false,securityDeposit=false,claimerBounty=false,usesNativeToken=
     true,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
77   PASS Valid claim (payOut=true,reputation=false,securityDeposit=false,claimerBounty=false,usesNativeToken=
     true,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
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     true,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
79   PASS Valid claim (payOut=true,reputation=true,securityDeposit=false,claimerBounty=false,usesNativeToken=
     true,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
80   PASS Valid claim (payOut=false,reputation=false,securityDeposit=true,claimerBounty=false,usesNativeToken=
     true,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
81   PASS Valid claim (payOut=true,reputation=false,securityDeposit=true,claimerBounty=false,usesNativeToken=
     true,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
82   PASS Valid claim (payOut=false,reputation=true,securityDeposit=true,claimerBounty=false,usesNativeToken=
     true,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
83   PASS Valid claim (payOut=true,reputation=true,securityDeposit=true,claimerBounty=false,usesNativeToken=true
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84   PASS Valid claim (payOut=false,reputation=false,securityDeposit=false,claimerBounty=true,usesNativeToken=
     true,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
85   PASS Valid claim (payOut=true,reputation=false,securityDeposit=false,claimerBounty=true,usesNativeToken=
     true,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
86   PASS Valid claim (payOut=false,reputation=true,securityDeposit=false,claimerBounty=true,usesNativeToken=
     true,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
87   PASS Valid claim (payOut=true,reputation=true,securityDeposit=false,claimerBounty=true,usesNativeToken=true
     ,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
88   PASS Valid claim (payOut=false,reputation=false,securityDeposit=true,claimerBounty=true,usesNativeToken=
     true,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
89   PASS Valid claim (payOut=true,reputation=false,securityDeposit=true,claimerBounty=true,usesNativeToken=true
     ,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
90   PASS Valid claim (payOut=false,reputation=true,securityDeposit=true,claimerBounty=true,usesNativeToken=true
     ,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
91   PASS Valid claim (payOut=true,reputation=true,securityDeposit=true,claimerBounty=true,usesNativeToken=true,
     usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=false)
92   PASS Valid claim (payOut=false,reputation=false,securityDeposit=false,claimerBounty=false,usesNativeToken=
     false,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
93   PASS Valid claim (payOut=true,reputation=false,securityDeposit=false,claimerBounty=false,usesNativeToken=
     false,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
94   PASS Valid claim (payOut=false,reputation=true,securityDeposit=false,claimerBounty=false,usesNativeToken=
     false,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
95   PASS Valid claim (payOut=true,reputation=true,securityDeposit=false,claimerBounty=false,usesNativeToken=
     false,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
96   PASS Valid claim (payOut=false,reputation=false,securityDeposit=true,claimerBounty=false,usesNativeToken=
     false,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
97   PASS Valid claim (payOut=true,reputation=false,securityDeposit=true,claimerBounty=false,usesNativeToken=
     false,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
98   PASS Valid claim (payOut=false,reputation=true,securityDeposit=true,claimerBounty=false,usesNativeToken=
     false,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
```

```
99    PASS Valid claim (payOut=true,reputation=true,securityDeposit=true,claimerBounty=false,usesNativeToken=
      false,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
100   PASS Valid claim (payOut=false,reputation=false,securityDeposit=false,claimerBounty=true,usesNativeToken=
      false,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
101   PASS Valid claim (payOut=true,reputation=false,securityDeposit=false,claimerBounty=true,usesNativeToken=
      false,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
102   PASS Valid claim (payOut=false,reputation=true,securityDeposit=false,claimerBounty=true,usesNativeToken=
      false,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
103   PASS Valid claim (payOut=true,reputation=true,securityDeposit=false,claimerBounty=true,usesNativeToken=
      false,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
104   PASS Valid claim (payOut=false,reputation=false,securityDeposit=true,claimerBounty=true,usesNativeToken=
      false,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
105   PASS Valid claim (payOut=true,reputation=false,securityDeposit=true,claimerBounty=true,usesNativeToken=
      false,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
106   PASS Valid claim (payOut=false,reputation=true,securityDeposit=true,claimerBounty=true,usesNativeToken=
      false,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
107   PASS Valid claim (payOut=true,reputation=true,securityDeposit=true,claimerBounty=true,usesNativeToken=false
      ,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
108   PASS Valid claim (payOut=false,reputation=false,securityDeposit=false,claimerBounty=false,usesNativeToken=
      true,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
109   PASS Valid claim (payOut=true,reputation=false,securityDeposit=false,claimerBounty=false,usesNativeToken=
      true,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
110   PASS Valid claim (payOut=false,reputation=true,securityDeposit=false,claimerBounty=false,usesNativeToken=
      true,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
111   PASS Valid claim (payOut=true,reputation=true,securityDeposit=false,claimerBounty=false,usesNativeToken=
      true,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
112   PASS Valid claim (payOut=false,reputation=false,securityDeposit=true,claimerBounty=false,usesNativeToken=
      true,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
113   PASS Valid claim (payOut=true,reputation=false,securityDeposit=true,claimerBounty=false,usesNativeToken=
      true,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
114   PASS Valid claim (payOut=false,reputation=true,securityDeposit=true,claimerBounty=false,usesNativeToken=
      true,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
115   PASS Valid claim (payOut=true,reputation=true,securityDeposit=true,claimerBounty=false,usesNativeToken=true
      ,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
116   PASS Valid claim (payOut=false,reputation=false,securityDeposit=false,claimerBounty=true,usesNativeToken=
      true,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
117   PASS Valid claim (payOut=true,reputation=false,securityDeposit=false,claimerBounty=true,usesNativeToken=
      true,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
118   PASS Valid claim (payOut=false,reputation=true,securityDeposit=false,claimerBounty=true,usesNativeToken=
      true,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
119   PASS Valid claim (payOut=true,reputation=true,securityDeposit=false,claimerBounty=true,usesNativeToken=true
      ,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
120   PASS Valid claim (payOut=false,reputation=false,securityDeposit=true,claimerBounty=true,usesNativeToken=
      true,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
121   PASS Valid claim (payOut=true,reputation=false,securityDeposit=true,claimerBounty=true,usesNativeToken=true
      ,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
122   PASS Valid claim (payOut=false,reputation=true,securityDeposit=true,claimerBounty=true,usesNativeToken=true
      ,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
123   PASS Valid claim (payOut=true,reputation=true,securityDeposit=true,claimerBounty=true,usesNativeToken=true,
      usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=false)
124   PASS Valid claim (payOut=false,reputation=false,securityDeposit=false,claimerBounty=false,usesNativeToken=
      false,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
125   PASS Valid claim (payOut=true,reputation=false,securityDeposit=false,claimerBounty=false,usesNativeToken=
      false,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
126   PASS Valid claim (payOut=false,reputation=true,securityDeposit=false,claimerBounty=false,usesNativeToken=
      false,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
127   PASS Valid claim (payOut=true,reputation=true,securityDeposit=false,claimerBounty=false,usesNativeToken=
      false,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
128   PASS Valid claim (payOut=false,reputation=false,securityDeposit=true,claimerBounty=false,usesNativeToken=
      false,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
129   PASS Valid claim (payOut=true,reputation=false,securityDeposit=true,claimerBounty=false,usesNativeToken=
      false,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
130   PASS Valid claim (payOut=false,reputation=true,securityDeposit=true,claimerBounty=false,usesNativeToken=
      false,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
131   PASS Valid claim (payOut=true,reputation=true,securityDeposit=true,claimerBounty=false,usesNativeToken=
      false,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
132   PASS Valid claim (payOut=false,reputation=false,securityDeposit=false,claimerBounty=true,usesNativeToken=
      false,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
133   PASS Valid claim (payOut=true,reputation=false,securityDeposit=false,claimerBounty=true,usesNativeToken=
      false,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
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134   PASS Valid claim (payOut=false,reputation=true,securityDeposit=false,claimerBounty=true,usesNativeToken=
      false,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
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      false,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
136   PASS Valid claim (payOut=false,reputation=false,securityDeposit=true,claimerBounty=true,usesNativeToken=
      false,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
137   PASS Valid claim (payOut=true,reputation=false,securityDeposit=true,claimerBounty=true,usesNativeToken=
      false,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
138   PASS Valid claim (payOut=false,reputation=true,securityDeposit=true,claimerBounty=true,usesNativeToken=
      false,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
139   PASS Valid claim (payOut=true,reputation=true,securityDeposit=true,claimerBounty=true,usesNativeToken=false
      ,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
140   PASS Valid claim (payOut=false,reputation=false,securityDeposit=false,claimerBounty=false,usesNativeToken=
      true,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
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      true,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
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      true,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
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      true,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
144   PASS Valid claim (payOut=false,reputation=false,securityDeposit=true,claimerBounty=false,usesNativeToken=
      true,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
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      true,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
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      true,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
147   PASS Valid claim (payOut=true,reputation=true,securityDeposit=true,claimerBounty=false,usesNativeToken=true
      ,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
148   PASS Valid claim (payOut=false,reputation=false,securityDeposit=false,claimerBounty=true,usesNativeToken=
      true,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
149   PASS Valid claim (payOut=true,reputation=false,securityDeposit=false,claimerBounty=true,usesNativeToken=
      true,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
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      true,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
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      true,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
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      ,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
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      ,usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
155   PASS Valid claim (payOut=true,reputation=true,securityDeposit=true,claimerBounty=true,usesNativeToken=true,
      usesNativeTokenForDeposit=false,successActionSuccess=true,successActionError=false)
156   PASS Valid claim (payOut=false,reputation=false,securityDeposit=false,claimerBounty=false,usesNativeToken=
      false,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
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      false,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
158   PASS Valid claim (payOut=false,reputation=true,securityDeposit=false,claimerBounty=false,usesNativeToken=
      false,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
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      false,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
160   PASS Valid claim (payOut=false,reputation=false,securityDeposit=true,claimerBounty=false,usesNativeToken=
      false,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
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      false,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
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      false,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
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      false,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
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      false,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
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      false,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
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      false,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
167   PASS Valid claim (payOut=true,reputation=true,securityDeposit=false,claimerBounty=true,usesNativeToken=
      false,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
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      false,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
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169       PASS Valid claim (payOut=true,reputation=false,securityDeposit=true,claimerBounty=true,usesNativeToken=
          false,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
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          false,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
171       PASS Valid claim (payOut=true,reputation=true,securityDeposit=true,claimerBounty=true,usesNativeToken=false
          ,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
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          true,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
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          true,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
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          true,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
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          true,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
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          true,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
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          ,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
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          true,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
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          true,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
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          true,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
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          ,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
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          true,usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
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          usesNativeTokenForDeposit=true,successActionSuccess=true,successActionError=false)
188       PASS Valid claim (payOut=false,reputation=false,securityDeposit=false,claimerBounty=false,usesNativeToken=
          false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
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          false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
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          false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
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          false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
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          false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
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          false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
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          false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
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          false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
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          false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
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          false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
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          false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
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          false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
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          false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
202       PASS Valid claim (payOut=false,reputation=true,securityDeposit=true,claimerBounty=true,usesNativeToken=
          false,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
203       PASS Valid claim (payOut=true,reputation=true,securityDeposit=true,claimerBounty=true,usesNativeToken=false
          ,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
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204   PASS Valid claim (payOut=false,reputation=false,securityDeposit=false,claimerBounty=false,usesNativeToken=
      true,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
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      true,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
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      true,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
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      true,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
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      true,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
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      true,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
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      true,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
211   PASS Valid claim (payOut=true,reputation=true,securityDeposit=true,claimerBounty=false,usesNativeToken=true
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      true,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
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      true,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
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      true,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
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      true,usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
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      usesNativeTokenForDeposit=false,successActionSuccess=false,successActionError=true)
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      false,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=true)
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      false,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=true)
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      false,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=true)
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      false,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=true)
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235   PASS Valid claim (payOut=true,reputation=true,securityDeposit=true,claimerBounty=true,usesNativeToken=false
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      true,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=true)
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      true,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=true)
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  true,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=true)
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  true,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=true)
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  true,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=true)
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  true,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=true)
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    PASS Valid claim (payOut=false,reputation=false,securityDeposit=false,claimerBounty=true,usesNativeToken=
  true,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=true)
    PASS Valid claim (payOut=true,reputation=false,securityDeposit=false,claimerBounty=true,usesNativeToken=
  true,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=true)
    PASS Valid claim (payOut=false,reputation=true,securityDeposit=false,claimerBounty=true,usesNativeToken=
  true,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=true)
    PASS Valid claim (payOut=true,reputation=true,securityDeposit=false,claimerBounty=true,usesNativeToken=true
  ,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=true)
    PASS Valid claim (payOut=false,reputation=false,securityDeposit=true,claimerBounty=true,usesNativeToken=
  true,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=true)
    PASS Valid claim (payOut=true,reputation=false,securityDeposit=true,claimerBounty=true,usesNativeToken=true
  ,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=true)
    PASS Valid claim (payOut=false,reputation=true,securityDeposit=true,claimerBounty=true,usesNativeToken=true
  ,usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=true)
    PASS Valid claim (payOut=true,reputation=true,securityDeposit=true,claimerBounty=true,usesNativeToken=true,
  usesNativeTokenForDeposit=true,successActionSuccess=false,successActionError=true)
    PASS Valid claim (contract call runs out of gas)
    PASS Invalid claim (negative answer from claim handler)
    PASS Invalid claim (claim uninitialized escrow)
    PASS Invalid claim (claim twice)
    PASS Invalid claim (no execution specified, but success action execution attempted)
    PASS Invalid claim (success action execution specified, but no success action execution attempted)
  Refund cooperative
    PASS Valid cooperative refund (payIn=false,reputation=false,securityDeposit=false,claimerBounty=false,
  usesNativeToken=false,usesNativeTokenForDeposit=false)
    PASS Valid cooperative refund (payIn=true,reputation=false,securityDeposit=false,claimerBounty=false,
  usesNativeToken=false,usesNativeTokenForDeposit=false)
    PASS Valid cooperative refund (payIn=false,reputation=true,securityDeposit=false,claimerBounty=false,
  usesNativeToken=false,usesNativeTokenForDeposit=false)
    PASS Valid cooperative refund (payIn=true,reputation=true,securityDeposit=false,claimerBounty=false,
  usesNativeToken=false,usesNativeTokenForDeposit=false)
    PASS Valid cooperative refund (payIn=false,reputation=false,securityDeposit=true,claimerBounty=false,
  usesNativeToken=false,usesNativeTokenForDeposit=false)
    PASS Valid cooperative refund (payIn=true,reputation=false,securityDeposit=true,claimerBounty=false,
  usesNativeToken=false,usesNativeTokenForDeposit=false)
    PASS Valid cooperative refund (payIn=false,reputation=true,securityDeposit=true,claimerBounty=false,
  usesNativeToken=false,usesNativeTokenForDeposit=false)
    PASS Valid cooperative refund (payIn=true,reputation=true,securityDeposit=true,claimerBounty=false,
  usesNativeToken=false,usesNativeTokenForDeposit=false)
    PASS Valid cooperative refund (payIn=false,reputation=false,securityDeposit=false,claimerBounty=true,
  usesNativeToken=false,usesNativeTokenForDeposit=false)
    PASS Valid cooperative refund (payIn=true,reputation=false,securityDeposit=false,claimerBounty=true,
  usesNativeToken=false,usesNativeTokenForDeposit=false)
    PASS Valid cooperative refund (payIn=false,reputation=true,securityDeposit=false,claimerBounty=true,
  usesNativeToken=false,usesNativeTokenForDeposit=false)
    PASS Valid cooperative refund (payIn=true,reputation=true,securityDeposit=false,claimerBounty=true,
  usesNativeToken=false,usesNativeTokenForDeposit=false)
    PASS Valid cooperative refund (payIn=false,reputation=false,securityDeposit=true,claimerBounty=true,
  usesNativeToken=false,usesNativeTokenForDeposit=false)
    PASS Valid cooperative refund (payIn=true,reputation=false,securityDeposit=true,claimerBounty=true,
  usesNativeToken=false,usesNativeTokenForDeposit=false)
    PASS Valid cooperative refund (payIn=false,reputation=true,securityDeposit=true,claimerBounty=true,
  usesNativeToken=false,usesNativeTokenForDeposit=false)
    PASS Valid cooperative refund (payIn=true,reputation=true,securityDeposit=true,claimerBounty=true,
  usesNativeToken=false,usesNativeTokenForDeposit=false)
    PASS Valid cooperative refund (payIn=false,reputation=false,securityDeposit=false,claimerBounty=false,
  usesNativeToken=true,usesNativeTokenForDeposit=false)
    PASS Valid cooperative refund (payIn=true,reputation=false,securityDeposit=false,claimerBounty=false,
  usesNativeToken=true,usesNativeTokenForDeposit=false)
```

```
277     PASS Valid cooperative refund (payIn=false,reputation=true,securityDeposit=false,claimerBounty=false,
        usesNativeToken=true,usesNativeTokenForDeposit=false)
278     PASS Valid cooperative refund (payIn=true,reputation=true,securityDeposit=false,claimerBounty=false,
        usesNativeToken=true,usesNativeTokenForDeposit=false)
279     PASS Valid cooperative refund (payIn=false,reputation=false,securityDeposit=true,claimerBounty=false,
        usesNativeToken=true,usesNativeTokenForDeposit=false)
280     PASS Valid cooperative refund (payIn=true,reputation=false,securityDeposit=true,claimerBounty=false,
        usesNativeToken=true,usesNativeTokenForDeposit=false)
281     PASS Valid cooperative refund (payIn=false,reputation=true,securityDeposit=true,claimerBounty=false,
        usesNativeToken=true,usesNativeTokenForDeposit=false)
282     PASS Valid cooperative refund (payIn=true,reputation=true,securityDeposit=true,claimerBounty=false,
        usesNativeToken=true,usesNativeTokenForDeposit=false)
283     PASS Valid cooperative refund (payIn=false,reputation=false,securityDeposit=false,claimerBounty=true,
        usesNativeToken=true,usesNativeTokenForDeposit=false)
284     PASS Valid cooperative refund (payIn=true,reputation=false,securityDeposit=false,claimerBounty=true,
        usesNativeToken=true,usesNativeTokenForDeposit=false)
285     PASS Valid cooperative refund (payIn=false,reputation=true,securityDeposit=false,claimerBounty=true,
        usesNativeToken=true,usesNativeTokenForDeposit=false)
286     PASS Valid cooperative refund (payIn=true,reputation=true,securityDeposit=false,claimerBounty=true,
        usesNativeToken=true,usesNativeTokenForDeposit=false)
287     PASS Valid cooperative refund (payIn=false,reputation=false,securityDeposit=true,claimerBounty=true,
        usesNativeToken=true,usesNativeTokenForDeposit=false)
288     PASS Valid cooperative refund (payIn=true,reputation=false,securityDeposit=true,claimerBounty=true,
        usesNativeToken=true,usesNativeTokenForDeposit=false)
289     PASS Valid cooperative refund (payIn=false,reputation=true,securityDeposit=true,claimerBounty=true,
        usesNativeToken=true,usesNativeTokenForDeposit=false)
290     PASS Valid cooperative refund (payIn=true,reputation=true,securityDeposit=true,claimerBounty=true,
        usesNativeToken=true,usesNativeTokenForDeposit=false)
291     PASS Valid cooperative refund (payIn=false,reputation=false,securityDeposit=false,claimerBounty=false,
        usesNativeToken=false,usesNativeTokenForDeposit=true)
292     PASS Valid cooperative refund (payIn=true,reputation=false,securityDeposit=false,claimerBounty=false,
        usesNativeToken=false,usesNativeTokenForDeposit=true)
293     PASS Valid cooperative refund (payIn=false,reputation=true,securityDeposit=false,claimerBounty=false,
        usesNativeToken=false,usesNativeTokenForDeposit=true)
294     PASS Valid cooperative refund (payIn=true,reputation=true,securityDeposit=false,claimerBounty=false,
        usesNativeToken=false,usesNativeTokenForDeposit=true)
295     PASS Valid cooperative refund (payIn=false,reputation=false,securityDeposit=true,claimerBounty=false,
        usesNativeToken=false,usesNativeTokenForDeposit=true)
296     PASS Valid cooperative refund (payIn=true,reputation=false,securityDeposit=true,claimerBounty=false,
        usesNativeToken=false,usesNativeTokenForDeposit=true)
297     PASS Valid cooperative refund (payIn=false,reputation=true,securityDeposit=true,claimerBounty=false,
        usesNativeToken=false,usesNativeTokenForDeposit=true)
298     PASS Valid cooperative refund (payIn=true,reputation=true,securityDeposit=true,claimerBounty=false,
        usesNativeToken=false,usesNativeTokenForDeposit=true)
299     PASS Valid cooperative refund (payIn=false,reputation=false,securityDeposit=false,claimerBounty=true,
        usesNativeToken=false,usesNativeTokenForDeposit=true)
300     PASS Valid cooperative refund (payIn=true,reputation=false,securityDeposit=false,claimerBounty=true,
        usesNativeToken=false,usesNativeTokenForDeposit=true)
301     PASS Valid cooperative refund (payIn=false,reputation=true,securityDeposit=false,claimerBounty=true,
        usesNativeToken=false,usesNativeTokenForDeposit=true)
302     PASS Valid cooperative refund (payIn=true,reputation=true,securityDeposit=false,claimerBounty=true,
        usesNativeToken=false,usesNativeTokenForDeposit=true)
303     PASS Valid cooperative refund (payIn=false,reputation=false,securityDeposit=true,claimerBounty=true,
        usesNativeToken=false,usesNativeTokenForDeposit=true)
304     PASS Valid cooperative refund (payIn=true,reputation=false,securityDeposit=true,claimerBounty=true,
        usesNativeToken=false,usesNativeTokenForDeposit=true)
305     PASS Valid cooperative refund (payIn=false,reputation=true,securityDeposit=true,claimerBounty=true,
        usesNativeToken=false,usesNativeTokenForDeposit=true)
306     PASS Valid cooperative refund (payIn=true,reputation=true,securityDeposit=true,claimerBounty=true,
        usesNativeToken=false,usesNativeTokenForDeposit=true)
307     PASS Valid cooperative refund (payIn=false,reputation=false,securityDeposit=false,claimerBounty=false,
        usesNativeToken=true,usesNativeTokenForDeposit=true)
308     PASS Valid cooperative refund (payIn=true,reputation=false,securityDeposit=false,claimerBounty=false,
        usesNativeToken=true,usesNativeTokenForDeposit=true)
309     PASS Valid cooperative refund (payIn=false,reputation=true,securityDeposit=false,claimerBounty=false,
        usesNativeToken=true,usesNativeTokenForDeposit=true)
310     PASS Valid cooperative refund (payIn=true,reputation=true,securityDeposit=false,claimerBounty=false,
        usesNativeToken=true,usesNativeTokenForDeposit=true)
311     PASS Valid cooperative refund (payIn=false,reputation=false,securityDeposit=true,claimerBounty=false,
        usesNativeToken=true,usesNativeTokenForDeposit=true)
```

```
      PASS Valid cooperative refund (payIn=true,reputation=false,securityDeposit=true,claimerBounty=false,
    usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid cooperative refund (payIn=false,reputation=true,securityDeposit=true,claimerBounty=false,
    usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid cooperative refund (payIn=true,reputation=true,securityDeposit=true,claimerBounty=false,
    usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid cooperative refund (payIn=false,reputation=false,securityDeposit=false,claimerBounty=true,
    usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid cooperative refund (payIn=true,reputation=false,securityDeposit=false,claimerBounty=true,
    usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid cooperative refund (payIn=false,reputation=true,securityDeposit=false,claimerBounty=true,
    usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid cooperative refund (payIn=true,reputation=true,securityDeposit=false,claimerBounty=true,
    usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid cooperative refund (payIn=false,reputation=false,securityDeposit=true,claimerBounty=true,
    usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid cooperative refund (payIn=true,reputation=false,securityDeposit=true,claimerBounty=true,
    usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid cooperative refund (payIn=false,reputation=true,securityDeposit=true,claimerBounty=true,
    usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid cooperative refund (payIn=true,reputation=true,securityDeposit=true,claimerBounty=true,
    usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid refund cooperative (claimer is an erc1271 account)
      PASS Invalid refund cooperative (not initialized)
      PASS Invalid refund cooperative (try to refund twice)
      PASS Invalid refund cooperative (timed out refund authorization)
      PASS Invalid refund cooperative (sign different timeout)
      PASS Invalid refund cooperative (sign random swap hash)
      PASS Invalid refund cooperative (wrong signer)
    Refund
      PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=false,
    securityDeposit=false,claimerBounty=false,usesNativeToken=false,usesNativeTokenForDeposit=false)
      PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=false,securityDeposit
    =false,claimerBounty=false,usesNativeToken=false,usesNativeTokenForDeposit=false)
      PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=true,securityDeposit
    =false,claimerBounty=false,usesNativeToken=false,usesNativeTokenForDeposit=false)
      PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=true,securityDeposit=
    false,claimerBounty=false,usesNativeToken=false,usesNativeTokenForDeposit=false)
      PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=false,
    securityDeposit=true,claimerBounty=false,usesNativeToken=false,usesNativeTokenForDeposit=false)
      PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=false,reputation=false,securityDeposit
    =true,claimerBounty=false,usesNativeToken=false,usesNativeTokenForDeposit=false)
      PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=false,securityDeposit
    =true,claimerBounty=false,usesNativeToken=false,usesNativeTokenForDeposit=false)
      PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=true,reputation=false,securityDeposit=
    true,claimerBounty=false,usesNativeToken=false,usesNativeTokenForDeposit=false)
      PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=true,securityDeposit
    =true,claimerBounty=false,usesNativeToken=false,usesNativeTokenForDeposit=false)
      PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=false,reputation=true,securityDeposit=
    true,claimerBounty=false,usesNativeToken=false,usesNativeTokenForDeposit=false)
      PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=true,securityDeposit=
    true,claimerBounty=false,usesNativeToken=false,usesNativeTokenForDeposit=false)
      PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=true,reputation=true,securityDeposit=
    true,claimerBounty=false,usesNativeToken=false,usesNativeTokenForDeposit=false)
      PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=false,
    securityDeposit=false,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=false)
      PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=false,reputation=false,securityDeposit
    =false,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=false)
      PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=false,securityDeposit
    =false,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=false)
      PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=true,reputation=false,securityDeposit=
    false,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=false)
      PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=true,securityDeposit
    =false,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=false)
      PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=false,reputation=true,securityDeposit=
    false,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=false)
      PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=true,securityDeposit=
    false,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=false)
      PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=true,reputation=true,securityDeposit=
    false,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=false)
```

```
351     PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=false,
        securityDeposit=true,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=false)
352     PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=false,reputation=false,securityDeposit
        =true,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=false)
353     PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=false,securityDeposit
        =true,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=false)
354     PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=true,reputation=false,securityDeposit=
        true,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=false)
355     PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=true,securityDeposit
        =true,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=false)
356     PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=false,reputation=true,securityDeposit=
        true,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=false)
357     PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=true,securityDeposit=
        true,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=false)
358     PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=true,reputation=true,securityDeposit=
        true,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=false)
359     PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=false,
        securityDeposit=false,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=false)
360     PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=false,securityDeposit
        =false,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=false)
361     PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=true,securityDeposit
        =false,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=false)
362     PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=true,securityDeposit=
        false,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=false)
363     PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=false,
        securityDeposit=true,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=false)
364     PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=false,reputation=false,securityDeposit
        =true,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=false)
365     PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=false,securityDeposit
        =true,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=false)
366     PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=true,reputation=false,securityDeposit=
        true,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=false)
367     PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=true,securityDeposit
        =true,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=false)
368     PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=false,reputation=true,securityDeposit=
        true,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=false)
369     PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=true,securityDeposit=
        true,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=false)
370     PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=true,reputation=true,securityDeposit=
        true,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=false)
371     PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=false,
        securityDeposit=false,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=false)
372     PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=false,reputation=false,securityDeposit
        =false,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=false)
373     PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=false,securityDeposit
        =false,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=false)
374     PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=true,reputation=false,securityDeposit=
        false,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=false)
375     PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=true,securityDeposit
        =false,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=false)
376     PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=false,reputation=true,securityDeposit=
        false,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=false)
377     PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=true,securityDeposit=
        false,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=false)
378     PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=true,reputation=true,securityDeposit=
        false,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=false)
379     PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=false,
        securityDeposit=true,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=false)
380     PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=false,reputation=false,securityDeposit
        =true,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=false)
381     PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=false,securityDeposit
        =true,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=false)
382     PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=true,reputation=false,securityDeposit=
        true,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=false)
383     PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=true,securityDeposit
        =true,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=false)
384     PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=false,reputation=true,securityDeposit=
        true,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=false)
385     PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=true,securityDeposit=
        true,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=false)
```

```
  PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=true,reputation=true,securityDeposit=true,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=false)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=false,securityDeposit=false,claimerBounty=false,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=false,securityDeposit=false,claimerBounty=false,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=true,securityDeposit=false,claimerBounty=false,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=true,securityDeposit=false,claimerBounty=false,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=false,securityDeposit=true,claimerBounty=false,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=false,reputation=false,securityDeposit=true,claimerBounty=false,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=false,securityDeposit=true,claimerBounty=false,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=true,reputation=false,securityDeposit=true,claimerBounty=false,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=true,securityDeposit=true,claimerBounty=false,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=false,reputation=true,securityDeposit=true,claimerBounty=false,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=true,securityDeposit=true,claimerBounty=false,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=true,reputation=true,securityDeposit=true,claimerBounty=false,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=false,securityDeposit=false,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=false,reputation=false,securityDeposit=false,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=false,securityDeposit=false,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=true,reputation=false,securityDeposit=false,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=true,securityDeposit=false,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=false,reputation=true,securityDeposit=false,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=true,securityDeposit=false,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=true,reputation=true,securityDeposit=false,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=false,securityDeposit=true,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=false,reputation=false,securityDeposit=true,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=false,securityDeposit=true,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=true,reputation=false,securityDeposit=true,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=true,securityDeposit=true,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=false,reputation=true,securityDeposit=true,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=true,securityDeposit=true,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=true,reputation=true,securityDeposit=true,claimerBounty=true,usesNativeToken=false,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=false,securityDeposit=false,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=false,securityDeposit=false,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=true,securityDeposit=false,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=true,securityDeposit=false,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=false,securityDeposit=true,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=true)
  PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=false,reputation=false,securityDeposit=true,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=true)
```

```
421       PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=false,securityDeposit
      =true,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=true)
422       PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=true,reputation=false,securityDeposit=
      true,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=true)
423       PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=true,securityDeposit
      =true,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=true)
424       PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=false,reputation=true,securityDeposit=
      true,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=true)
425       PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=true,securityDeposit=
      true,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=true)
426       PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=true,reputation=true,securityDeposit=
      true,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=true)
427       PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=false,
      securityDeposit=false,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=true)
428       PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=false,reputation=false,securityDeposit
      =false,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=true)
429       PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=false,securityDeposit
      =false,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=true)
430       PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=true,reputation=false,securityDeposit=
      false,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=true)
431       PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=true,securityDeposit
      =false,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=true)
432       PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=false,reputation=true,securityDeposit=
      false,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=true)
433       PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=true,securityDeposit=
      false,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=true)
434       PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=true,reputation=true,securityDeposit=
      false,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=true)
435       PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=false,
      securityDeposit=true,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=true)
436       PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=false,reputation=false,securityDeposit
      =true,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=true)
437       PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=false,securityDeposit
      =true,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=true)
438       PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=true,reputation=false,securityDeposit=
      true,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=true)
439       PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=false,reputation=true,securityDeposit
      =true,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=true)
440       PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=false,reputation=true,securityDeposit=
      true,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=true)
441       PASS Valid refund (securityDepositLargerThanClaimerBounty=false,payIn=true,reputation=true,securityDeposit=
      true,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=true)
442       PASS Valid refund (securityDepositLargerThanClaimerBounty=true,payIn=true,reputation=true,securityDeposit=
      true,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=true)
443       PASS Invalid refund (negative answer from refund handler)
444       PASS Invalid refund (refund uninitialized escrow)
445       PASS Invalid refund (refund twice)
446     Initialize
447       PASS Valid initialize (senderClaimer=false,payIn=false,reputation=false,securityDeposit=false,claimerBounty
      =false,usesNativeToken=false,usesNativeTokenForDeposit=false)
448       PASS Valid initialize (senderClaimer=true,payIn=false,reputation=false,securityDeposit=false,claimerBounty=
      false,usesNativeToken=false,usesNativeTokenForDeposit=false)
449       PASS Valid initialize (senderClaimer=false,payIn=true,reputation=false,securityDeposit=false,claimerBounty=
      false,usesNativeToken=false,usesNativeTokenForDeposit=false)
450       PASS Valid initialize (senderClaimer=true,payIn=true,reputation=false,securityDeposit=false,claimerBounty=
      false,usesNativeToken=false,usesNativeTokenForDeposit=false)
451       PASS Valid initialize (senderClaimer=false,payIn=false,reputation=true,securityDeposit=false,claimerBounty=
      false,usesNativeToken=false,usesNativeTokenForDeposit=false)
452       PASS Valid initialize (senderClaimer=true,payIn=false,reputation=true,securityDeposit=false,claimerBounty=
      false,usesNativeToken=false,usesNativeTokenForDeposit=false)
453       PASS Valid initialize (senderClaimer=false,payIn=true,reputation=true,securityDeposit=false,claimerBounty=
      false,usesNativeToken=false,usesNativeTokenForDeposit=false)
454       PASS Valid initialize (senderClaimer=true,payIn=true,reputation=true,securityDeposit=false,claimerBounty=
      false,usesNativeToken=false,usesNativeTokenForDeposit=false)
455       PASS Valid initialize (senderClaimer=false,payIn=false,reputation=false,securityDeposit=true,claimerBounty=
      false,usesNativeToken=false,usesNativeTokenForDeposit=false)
456       PASS Valid initialize (senderClaimer=true,payIn=false,reputation=false,securityDeposit=true,claimerBounty=
      false,usesNativeToken=false,usesNativeTokenForDeposit=false)
457       PASS Valid initialize (senderClaimer=false,payIn=true,reputation=false,securityDeposit=true,claimerBounty=
      false,usesNativeToken=false,usesNativeTokenForDeposit=false)
```

```
458   PASS Valid initialize (senderClaimer=true,payIn=true,reputation=false,securityDeposit=true,claimerBounty=
      false,usesNativeToken=false,usesNativeTokenForDeposit=false)
459   PASS Valid initialize (senderClaimer=false,payIn=false,reputation=true,securityDeposit=true,claimerBounty=
      false,usesNativeToken=false,usesNativeTokenForDeposit=false)
460   PASS Valid initialize (senderClaimer=true,payIn=false,reputation=true,securityDeposit=true,claimerBounty=
      false,usesNativeToken=false,usesNativeTokenForDeposit=false)
461   PASS Valid initialize (senderClaimer=false,payIn=true,reputation=true,securityDeposit=true,claimerBounty=
      false,usesNativeToken=false,usesNativeTokenForDeposit=false)
462   PASS Valid initialize (senderClaimer=true,payIn=true,reputation=true,securityDeposit=true,claimerBounty=
      false,usesNativeToken=false,usesNativeTokenForDeposit=false)
463   PASS Valid initialize (senderClaimer=false,payIn=false,reputation=false,securityDeposit=false,claimerBounty
      =true,usesNativeToken=false,usesNativeTokenForDeposit=false)
464   PASS Valid initialize (senderClaimer=true,payIn=false,reputation=false,securityDeposit=false,claimerBounty=
      true,usesNativeToken=false,usesNativeTokenForDeposit=false)
465   PASS Valid initialize (senderClaimer=false,payIn=true,reputation=false,securityDeposit=false,claimerBounty=
      true,usesNativeToken=false,usesNativeTokenForDeposit=false)
466   PASS Valid initialize (senderClaimer=true,payIn=true,reputation=false,securityDeposit=false,claimerBounty=
      true,usesNativeToken=false,usesNativeTokenForDeposit=false)
467   PASS Valid initialize (senderClaimer=false,payIn=false,reputation=true,securityDeposit=false,claimerBounty=
      true,usesNativeToken=false,usesNativeTokenForDeposit=false)
468   PASS Valid initialize (senderClaimer=true,payIn=false,reputation=true,securityDeposit=false,claimerBounty=
      true,usesNativeToken=false,usesNativeTokenForDeposit=false)
469   PASS Valid initialize (senderClaimer=false,payIn=true,reputation=true,securityDeposit=false,claimerBounty=
      true,usesNativeToken=false,usesNativeTokenForDeposit=false)
470   PASS Valid initialize (senderClaimer=true,payIn=true,reputation=true,securityDeposit=false,claimerBounty=
      true,usesNativeToken=false,usesNativeTokenForDeposit=false)
471   PASS Valid initialize (senderClaimer=false,payIn=false,reputation=false,securityDeposit=true,claimerBounty=
      true,usesNativeToken=false,usesNativeTokenForDeposit=false)
472   PASS Valid initialize (senderClaimer=true,payIn=false,reputation=false,securityDeposit=true,claimerBounty=
      true,usesNativeToken=false,usesNativeTokenForDeposit=false)
473   PASS Valid initialize (senderClaimer=false,payIn=true,reputation=false,securityDeposit=true,claimerBounty=
      true,usesNativeToken=false,usesNativeTokenForDeposit=false)
474   PASS Valid initialize (senderClaimer=true,payIn=true,reputation=false,securityDeposit=true,claimerBounty=
      true,usesNativeToken=false,usesNativeTokenForDeposit=false)
475   PASS Valid initialize (senderClaimer=false,payIn=false,reputation=true,securityDeposit=true,claimerBounty=
      true,usesNativeToken=false,usesNativeTokenForDeposit=false)
476   PASS Valid initialize (senderClaimer=true,payIn=false,reputation=true,securityDeposit=true,claimerBounty=
      true,usesNativeToken=false,usesNativeTokenForDeposit=false)
477   PASS Valid initialize (senderClaimer=false,payIn=true,reputation=true,securityDeposit=true,claimerBounty=
      true,usesNativeToken=false,usesNativeTokenForDeposit=false)
478   PASS Valid initialize (senderClaimer=true,payIn=true,reputation=true,securityDeposit=true,claimerBounty=
      true,usesNativeToken=false,usesNativeTokenForDeposit=false)
479   PASS Valid initialize (senderClaimer=false,payIn=false,reputation=false,securityDeposit=false,claimerBounty
      =false,usesNativeToken=true,usesNativeTokenForDeposit=false)
480   PASS Valid initialize (senderClaimer=false,payIn=true,reputation=false,securityDeposit=false,claimerBounty=
      false,usesNativeToken=true,usesNativeTokenForDeposit=false)
481   PASS Valid initialize (senderClaimer=false,payIn=false,reputation=true,securityDeposit=false,claimerBounty=
      false,usesNativeToken=true,usesNativeTokenForDeposit=false)
482   PASS Valid initialize (senderClaimer=false,payIn=true,reputation=true,securityDeposit=false,claimerBounty=
      false,usesNativeToken=true,usesNativeTokenForDeposit=false)
483   PASS Valid initialize (senderClaimer=false,payIn=false,reputation=false,securityDeposit=true,claimerBounty=
      false,usesNativeToken=true,usesNativeTokenForDeposit=false)
484   PASS Valid initialize (senderClaimer=false,payIn=true,reputation=false,securityDeposit=true,claimerBounty=
      false,usesNativeToken=true,usesNativeTokenForDeposit=false)
485   PASS Valid initialize (senderClaimer=false,payIn=false,reputation=true,securityDeposit=true,claimerBounty=
      false,usesNativeToken=true,usesNativeTokenForDeposit=false)
486   PASS Valid initialize (senderClaimer=false,payIn=true,reputation=true,securityDeposit=true,claimerBounty=
      false,usesNativeToken=true,usesNativeTokenForDeposit=false)
487   PASS Valid initialize (senderClaimer=false,payIn=false,reputation=false,securityDeposit=false,claimerBounty
      =true,usesNativeToken=true,usesNativeTokenForDeposit=false)
488   PASS Valid initialize (senderClaimer=false,payIn=true,reputation=false,securityDeposit=false,claimerBounty=
      true,usesNativeToken=true,usesNativeTokenForDeposit=false)
489   PASS Valid initialize (senderClaimer=false,payIn=false,reputation=true,securityDeposit=false,claimerBounty=
      true,usesNativeToken=true,usesNativeTokenForDeposit=false)
490   PASS Valid initialize (senderClaimer=false,payIn=true,reputation=true,securityDeposit=false,claimerBounty=
      true,usesNativeToken=true,usesNativeTokenForDeposit=false)
491   PASS Valid initialize (senderClaimer=false,payIn=false,reputation=false,securityDeposit=true,claimerBounty=
      true,usesNativeToken=true,usesNativeTokenForDeposit=false)
492   PASS Valid initialize (senderClaimer=false,payIn=true,reputation=false,securityDeposit=true,claimerBounty=
      true,usesNativeToken=true,usesNativeTokenForDeposit=false)
```

```
493       PASS Valid initialize (senderClaimer=false,payIn=false,reputation=true,securityDeposit=true,claimerBounty=
          true,usesNativeToken=true,usesNativeTokenForDeposit=false)
494       PASS Valid initialize (senderClaimer=false,payIn=true,reputation=true,securityDeposit=true,claimerBounty=
          true,usesNativeToken=true,usesNativeTokenForDeposit=false)
495       PASS Valid initialize (senderClaimer=false,payIn=false,reputation=false,securityDeposit=false,claimerBounty
          =false,usesNativeToken=false,usesNativeTokenForDeposit=true)
496       PASS Valid initialize (senderClaimer=true,payIn=false,reputation=false,securityDeposit=false,claimerBounty=
          false,usesNativeToken=false,usesNativeTokenForDeposit=true)
497       PASS Valid initialize (senderClaimer=false,payIn=true,reputation=false,securityDeposit=false,claimerBounty=
          false,usesNativeToken=false,usesNativeTokenForDeposit=true)
498       PASS Valid initialize (senderClaimer=true,payIn=true,reputation=false,securityDeposit=false,claimerBounty=
          false,usesNativeToken=false,usesNativeTokenForDeposit=true)
499       PASS Valid initialize (senderClaimer=false,payIn=false,reputation=true,securityDeposit=false,claimerBounty=
          false,usesNativeToken=false,usesNativeTokenForDeposit=true)
500       PASS Valid initialize (senderClaimer=true,payIn=false,reputation=true,securityDeposit=false,claimerBounty=
          false,usesNativeToken=false,usesNativeTokenForDeposit=true)
501       PASS Valid initialize (senderClaimer=false,payIn=true,reputation=true,securityDeposit=false,claimerBounty=
          false,usesNativeToken=false,usesNativeTokenForDeposit=true)
502       PASS Valid initialize (senderClaimer=true,payIn=true,reputation=true,securityDeposit=false,claimerBounty=
          false,usesNativeToken=false,usesNativeTokenForDeposit=true)
503       PASS Valid initialize (senderClaimer=false,payIn=false,reputation=false,securityDeposit=true,claimerBounty=
          false,usesNativeToken=false,usesNativeTokenForDeposit=true)
504       PASS Valid initialize (senderClaimer=true,payIn=false,reputation=false,securityDeposit=true,claimerBounty=
          false,usesNativeToken=false,usesNativeTokenForDeposit=true)
505       PASS Valid initialize (senderClaimer=false,payIn=true,reputation=false,securityDeposit=true,claimerBounty=
          false,usesNativeToken=false,usesNativeTokenForDeposit=true)
506       PASS Valid initialize (senderClaimer=true,payIn=true,reputation=false,securityDeposit=true,claimerBounty=
          false,usesNativeToken=false,usesNativeTokenForDeposit=true)
507       PASS Valid initialize (senderClaimer=false,payIn=false,reputation=true,securityDeposit=true,claimerBounty=
          false,usesNativeToken=false,usesNativeTokenForDeposit=true)
508       PASS Valid initialize (senderClaimer=true,payIn=false,reputation=true,securityDeposit=true,claimerBounty=
          false,usesNativeToken=false,usesNativeTokenForDeposit=true)
509       PASS Valid initialize (senderClaimer=false,payIn=true,reputation=true,securityDeposit=true,claimerBounty=
          false,usesNativeToken=false,usesNativeTokenForDeposit=true)
510       PASS Valid initialize (senderClaimer=true,payIn=true,reputation=true,securityDeposit=true,claimerBounty=
          false,usesNativeToken=false,usesNativeTokenForDeposit=true)
511       PASS Valid initialize (senderClaimer=false,payIn=false,reputation=false,securityDeposit=false,claimerBounty
          =true,usesNativeToken=false,usesNativeTokenForDeposit=true)
512       PASS Valid initialize (senderClaimer=true,payIn=false,reputation=false,securityDeposit=false,claimerBounty=
          true,usesNativeToken=false,usesNativeTokenForDeposit=true)
513       PASS Valid initialize (senderClaimer=false,payIn=true,reputation=false,securityDeposit=false,claimerBounty=
          true,usesNativeToken=false,usesNativeTokenForDeposit=true)
514       PASS Valid initialize (senderClaimer=true,payIn=true,reputation=false,securityDeposit=false,claimerBounty=
          true,usesNativeToken=false,usesNativeTokenForDeposit=true)
515       PASS Valid initialize (senderClaimer=false,payIn=false,reputation=true,securityDeposit=false,claimerBounty=
          true,usesNativeToken=false,usesNativeTokenForDeposit=true)
516       PASS Valid initialize (senderClaimer=true,payIn=false,reputation=true,securityDeposit=false,claimerBounty=
          true,usesNativeToken=false,usesNativeTokenForDeposit=true)
517       PASS Valid initialize (senderClaimer=false,payIn=true,reputation=true,securityDeposit=false,claimerBounty=
          true,usesNativeToken=false,usesNativeTokenForDeposit=true)
518       PASS Valid initialize (senderClaimer=true,payIn=true,reputation=true,securityDeposit=false,claimerBounty=
          true,usesNativeToken=false,usesNativeTokenForDeposit=true)
519       PASS Valid initialize (senderClaimer=false,payIn=false,reputation=false,securityDeposit=true,claimerBounty=
          true,usesNativeToken=false,usesNativeTokenForDeposit=true)
520       PASS Valid initialize (senderClaimer=true,payIn=false,reputation=false,securityDeposit=true,claimerBounty=
          true,usesNativeToken=false,usesNativeTokenForDeposit=true)
521       PASS Valid initialize (senderClaimer=false,payIn=true,reputation=false,securityDeposit=true,claimerBounty=
          true,usesNativeToken=false,usesNativeTokenForDeposit=true)
522       PASS Valid initialize (senderClaimer=true,payIn=true,reputation=false,securityDeposit=true,claimerBounty=
          true,usesNativeToken=false,usesNativeTokenForDeposit=true)
523       PASS Valid initialize (senderClaimer=false,payIn=false,reputation=true,securityDeposit=true,claimerBounty=
          true,usesNativeToken=false,usesNativeTokenForDeposit=true)
524       PASS Valid initialize (senderClaimer=true,payIn=false,reputation=true,securityDeposit=true,claimerBounty=
          true,usesNativeToken=false,usesNativeTokenForDeposit=true)
525       PASS Valid initialize (senderClaimer=false,payIn=true,reputation=true,securityDeposit=true,claimerBounty=
          true,usesNativeToken=false,usesNativeTokenForDeposit=true)
526       PASS Valid initialize (senderClaimer=true,payIn=true,reputation=true,securityDeposit=true,claimerBounty=
          true,usesNativeToken=false,usesNativeTokenForDeposit=true)
527       PASS Valid initialize (senderClaimer=false,payIn=false,reputation=false,securityDeposit=false,claimerBounty
          =false,usesNativeToken=true,usesNativeTokenForDeposit=true)
```

```
      PASS Valid initialize (senderClaimer=false,payIn=true,reputation=false,securityDeposit=false,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid initialize (senderClaimer=false,payIn=false,reputation=true,securityDeposit=false,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid initialize (senderClaimer=false,payIn=true,reputation=true,securityDeposit=false,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid initialize (senderClaimer=false,payIn=false,reputation=false,securityDeposit=true,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid initialize (senderClaimer=false,payIn=true,reputation=false,securityDeposit=true,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid initialize (senderClaimer=false,payIn=false,reputation=true,securityDeposit=true,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid initialize (senderClaimer=false,payIn=true,reputation=true,securityDeposit=true,claimerBounty=false,usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid initialize (senderClaimer=false,payIn=false,reputation=false,securityDeposit=false,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid initialize (senderClaimer=false,payIn=true,reputation=false,securityDeposit=false,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid initialize (senderClaimer=false,payIn=false,reputation=true,securityDeposit=false,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid initialize (senderClaimer=false,payIn=true,reputation=true,securityDeposit=false,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid initialize (senderClaimer=false,payIn=false,reputation=false,securityDeposit=true,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid initialize (senderClaimer=false,payIn=true,reputation=false,securityDeposit=true,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid initialize (senderClaimer=false,payIn=false,reputation=true,securityDeposit=true,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid initialize (senderClaimer=false,payIn=true,reputation=true,securityDeposit=true,claimerBounty=true,usesNativeToken=true,usesNativeTokenForDeposit=true)
      PASS Valid initialize (claimer is an erc1271 account)
      PASS Invalid initialize not enough balance (payIn)
      PASS Invalid initialize not enough balance (not payIn)
      PASS Invalid initialize not enough native token sent to the contract
      PASS Invalid initialize (check correct handling of msg.value)
      PASS Invalid initialize not enough allowance (payIn erc-20)
      PASS Invalid initialize not enough deposit token balance (erc-20)
      PASS Invalid initialize not enough deposit token allowance (erc-20)
      PASS Invalid initialize wrong signer
      PASS Invalid initialize bad sign message
      PASS Invalid initialize bad sign message (extra data)
      PASS Invalid initialize 3rd party caller
      PASS Invalid initialize expired
      PASS Invalid initialize sign different timeout
      PASS Invalid initialize commit twice

  ExecutionContract
    PASS Valid create (erc-20) (40ms)
    PASS Invalid create (erc-20 not enough balance)
    PASS Invalid create (erc-20 not enough allowance)
    PASS Valid create (native token)
    PASS Invalid create (native token invalid msg.value)
    PASS Invalid create twice the same execution
    PASS Valid refund expired
    PASS Invalid refund expired, not initiated
    PASS Invalid refund expired, not expired
    PASS Invalid refund expired, already processed
    PASS Valid refund by owner
    PASS Valid refund by owner (even though not expired yet)
    PASS Invalid refund by owner, caller not owner
    PASS Invalid refund by owner, not initiated
    PASS Invalid refund by owner, try to refund twice
    PASS Valid execute, empty calls
    PASS Valid execute, erc20 transfer calls
    PASS Valid execute, native transfer calls
    PASS Valid execute, dummy emit event
    PASS Valid execute, call reverted
    PASS Valid execute, out of gas
    PASS Valid execute, dummy emit event (drain tokens)
    PASS Valid execute, call reverted (drain tokens)
```

```
    PASS Invalid execute, wrong success action provided
    PASS Invalid execute, not scheduled
    PASS Invalid execute, try execute twice

  SpvVaultManager
    Open
      PASS Valid open vault (60ms)
      PASS Invalid open vault (already opened)
    Deposit
      PASS Valid deposit (token0Native=false,token1Native=false,token0Type=false,token1Type=false,
     thirdPartyDeposit=false,noToken0=false,noToken1=false)
      PASS Valid deposit (token0Native=true,token1Native=false,token0Type=false,token1Type=false,
     thirdPartyDeposit=false,noToken0=false,noToken1=false)
      PASS Valid deposit (token0Native=false,token1Native=true,token0Type=false,token1Type=false,
     thirdPartyDeposit=false,noToken0=false,noToken1=false)
      PASS Valid deposit (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyDeposit
     =false,noToken0=false,noToken1=false)
      PASS Valid deposit (token0Native=false,token1Native=false,token0Type=true,token1Type=false,
     thirdPartyDeposit=false,noToken0=false,noToken1=false)
      PASS Valid deposit (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyDeposit
     =false,noToken0=false,noToken1=false)
      PASS Valid deposit (token0Native=false,token1Native=false,token0Type=false,token1Type=true,
     thirdPartyDeposit=false,noToken0=false,noToken1=false)
      PASS Valid deposit (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyDeposit
     =false,noToken0=false,noToken1=false)
      PASS Valid deposit (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyDeposit
     =false,noToken0=false,noToken1=false)
      PASS Valid deposit (token0Native=false,token1Native=false,token0Type=false,token1Type=false,
     thirdPartyDeposit=true,noToken0=false,noToken1=false)
      PASS Valid deposit (token0Native=true,token1Native=false,token0Type=false,token1Type=false,
     thirdPartyDeposit=true,noToken0=false,noToken1=false)
      PASS Valid deposit (token0Native=false,token1Native=true,token0Type=false,token1Type=false,
     thirdPartyDeposit=true,noToken0=false,noToken1=false)
      PASS Valid deposit (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyDeposit
     =true,noToken0=false,noToken1=false)
      PASS Valid deposit (token0Native=false,token1Native=false,token0Type=true,token1Type=false,
     thirdPartyDeposit=true,noToken0=false,noToken1=false)
      PASS Valid deposit (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyDeposit
     =true,noToken0=false,noToken1=false)
      PASS Valid deposit (token0Native=false,token1Native=false,token0Type=false,token1Type=true,
     thirdPartyDeposit=true,noToken0=false,noToken1=false)
      PASS Valid deposit (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyDeposit
     =true,noToken0=false,noToken1=false)
      PASS Valid deposit (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyDeposit
     =true,noToken0=false,noToken1=false)
      PASS Valid deposit (token0Native=false,token1Native=false,token0Type=false,token1Type=false,
     thirdPartyDeposit=false,noToken0=true,noToken1=false)
      PASS Valid deposit (token0Native=true,token1Native=false,token0Type=false,token1Type=false,
     thirdPartyDeposit=false,noToken0=true,noToken1=false)
      PASS Valid deposit (token0Native=false,token1Native=true,token0Type=false,token1Type=false,
     thirdPartyDeposit=false,noToken0=true,noToken1=false)
      PASS Valid deposit (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyDeposit
     =false,noToken0=true,noToken1=false)
      PASS Valid deposit (token0Native=false,token1Native=false,token0Type=true,token1Type=false,
     thirdPartyDeposit=false,noToken0=true,noToken1=false)
      PASS Valid deposit (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyDeposit
     =false,noToken0=true,noToken1=false)
      PASS Valid deposit (token0Native=false,token1Native=false,token0Type=false,token1Type=true,
     thirdPartyDeposit=false,noToken0=true,noToken1=false)
      PASS Valid deposit (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyDeposit
     =false,noToken0=true,noToken1=false)
      PASS Valid deposit (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyDeposit
     =false,noToken0=true,noToken1=false)
      PASS Valid deposit (token0Native=false,token1Native=false,token0Type=false,token1Type=false,
     thirdPartyDeposit=true,noToken0=true,noToken1=false)
      PASS Valid deposit (token0Native=true,token1Native=false,token0Type=false,token1Type=false,
     thirdPartyDeposit=true,noToken0=true,noToken1=false)
      PASS Valid deposit (token0Native=false,token1Native=true,token0Type=false,token1Type=false,
     thirdPartyDeposit=true,noToken0=true,noToken1=false)
```

```
622       PASS Valid deposit (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyDeposit
      =true,noToken0=true,noToken1=false)
623       PASS Valid deposit (token0Native=false,token1Native=false,token0Type=true,token1Type=false,
      thirdPartyDeposit=true,noToken0=true,noToken1=false)
624       PASS Valid deposit (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyDeposit
      =true,noToken0=true,noToken1=false)
625       PASS Valid deposit (token0Native=false,token1Native=false,token0Type=false,token1Type=true,
      thirdPartyDeposit=true,noToken0=true,noToken1=false)
626       PASS Valid deposit (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyDeposit
      =true,noToken0=true,noToken1=false)
627       PASS Valid deposit (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyDeposit
      =true,noToken0=true,noToken1=false)
628       PASS Valid deposit (token0Native=false,token1Native=false,token0Type=false,token1Type=false,
      thirdPartyDeposit=false,noToken0=false,noToken1=true)
629       PASS Valid deposit (token0Native=true,token1Native=false,token0Type=false,token1Type=false,
      thirdPartyDeposit=false,noToken0=false,noToken1=true)
630       PASS Valid deposit (token0Native=false,token1Native=true,token0Type=false,token1Type=false,
      thirdPartyDeposit=false,noToken0=false,noToken1=true)
631       PASS Valid deposit (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyDeposit
      =false,noToken0=false,noToken1=true)
632       PASS Valid deposit (token0Native=false,token1Native=false,token0Type=true,token1Type=false,
      thirdPartyDeposit=false,noToken0=false,noToken1=true)
633       PASS Valid deposit (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyDeposit
      =false,noToken0=false,noToken1=true)
634       PASS Valid deposit (token0Native=false,token1Native=false,token0Type=false,token1Type=true,
      thirdPartyDeposit=false,noToken0=false,noToken1=true)
635       PASS Valid deposit (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyDeposit
      =false,noToken0=false,noToken1=true)
636       PASS Valid deposit (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyDeposit
      =false,noToken0=false,noToken1=true)
637       PASS Valid deposit (token0Native=false,token1Native=false,token0Type=false,token1Type=false,
      thirdPartyDeposit=true,noToken0=false,noToken1=true)
638       PASS Valid deposit (token0Native=true,token1Native=false,token0Type=false,token1Type=false,
      thirdPartyDeposit=true,noToken0=false,noToken1=true)
639       PASS Valid deposit (token0Native=false,token1Native=true,token0Type=false,token1Type=false,
      thirdPartyDeposit=true,noToken0=false,noToken1=true)
640       PASS Valid deposit (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyDeposit
      =true,noToken0=false,noToken1=true)
641       PASS Valid deposit (token0Native=false,token1Native=false,token0Type=true,token1Type=false,
      thirdPartyDeposit=true,noToken0=false,noToken1=true)
642       PASS Valid deposit (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyDeposit
      =true,noToken0=false,noToken1=true)
643       PASS Valid deposit (token0Native=false,token1Native=false,token0Type=false,token1Type=true,
      thirdPartyDeposit=true,noToken0=false,noToken1=true)
644       PASS Valid deposit (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyDeposit
      =true,noToken0=false,noToken1=true)
645       PASS Valid deposit (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyDeposit
      =true,noToken0=false,noToken1=true)
646       PASS Invalid deposit - not enough balance erc-20 (token0Native=false,token1Native=false,token0Type=false,
      token1Type=false,token0NotEnoughBalance=true,token1NotEnoughBalance=false)
647       PASS Invalid deposit - not enough balance erc-20 (token0Native=false,token1Native=true,token0Type=false,
      token1Type=false,token0NotEnoughBalance=true,token1NotEnoughBalance=false)
648       PASS Invalid deposit - not enough balance erc-20 (token0Native=false,token1Native=false,token0Type=true,
      token1Type=false,token0NotEnoughBalance=true,token1NotEnoughBalance=false)
649       PASS Invalid deposit - not enough balance erc-20 (token0Native=false,token1Native=true,token0Type=true,
      token1Type=false,token0NotEnoughBalance=true,token1NotEnoughBalance=false)
650       PASS Invalid deposit - not enough balance erc-20 (token0Native=false,token1Native=false,token0Type=false,
      token1Type=true,token0NotEnoughBalance=true,token1NotEnoughBalance=false)
651       PASS Invalid deposit - not enough balance erc-20 (token0Native=false,token1Native=false,token0Type=true,
      token1Type=true,token0NotEnoughBalance=true,token1NotEnoughBalance=false)
652       PASS Invalid deposit - not enough balance erc-20 (token0Native=false,token1Native=false,token0Type=false,
      token1Type=false,token0NotEnoughBalance=false,token1NotEnoughBalance=true)
653       PASS Invalid deposit - not enough balance erc-20 (token0Native=true,token1Native=false,token0Type=false,
      token1Type=false,token0NotEnoughBalance=false,token1NotEnoughBalance=true)
654       PASS Invalid deposit - not enough balance erc-20 (token0Native=false,token1Native=false,token0Type=true,
      token1Type=false,token0NotEnoughBalance=false,token1NotEnoughBalance=true)
655       PASS Invalid deposit - not enough balance erc-20 (token0Native=false,token1Native=false,token0Type=false,
      token1Type=true,token0NotEnoughBalance=false,token1NotEnoughBalance=true)
656       PASS Invalid deposit - not enough balance erc-20 (token0Native=true,token1Native=false,token0Type=false,
      token1Type=true,token0NotEnoughBalance=false,token1NotEnoughBalance=true)
```

```
    PASS Invalid deposit - not enough balance erc-20 (token0Native=false,token1Native=false,token0Type=true,
    token1Type=true,token0NotEnoughBalance=false,token1NotEnoughBalance=true)
    PASS Invalid deposit - not enough balance erc-20 (token0Native=false,token1Native=false,token0Type=false,
    token1Type=false,token0NotEnoughBalance=true,token1NotEnoughBalance=true)
    PASS Invalid deposit - not enough balance erc-20 (token0Native=false,token1Native=false,token0Type=true,
    token1Type=false,token0NotEnoughBalance=true,token1NotEnoughBalance=true)
    PASS Invalid deposit - not enough balance erc-20 (token0Native=false,token1Native=false,token0Type=false,
    token1Type=true,token0NotEnoughBalance=true,token1NotEnoughBalance=true)
    PASS Invalid deposit - not enough balance erc-20 (token0Native=false,token1Native=false,token0Type=true,
    token1Type=true,token0NotEnoughBalance=true,token1NotEnoughBalance=true)
    PASS Invalid deposit - not enough allowance erc-20 (token0Native=false,token1Native=false,token0Type=false,
    token1Type=false,token0NotEnoughAllowance=true,token1NotEnoughAllowance=false)
    PASS Invalid deposit - not enough allowance erc-20 (token0Native=false,token1Native=true,token0Type=false,
    token1Type=false,token0NotEnoughAllowance=true,token1NotEnoughAllowance=false)
    PASS Invalid deposit - not enough allowance erc-20 (token0Native=false,token1Native=false,token0Type=true,
    token1Type=false,token0NotEnoughAllowance=true,token1NotEnoughAllowance=false)
    PASS Invalid deposit - not enough allowance erc-20 (token0Native=false,token1Native=true,token0Type=true,
    token1Type=false,token0NotEnoughAllowance=true,token1NotEnoughAllowance=false)
    PASS Invalid deposit - not enough allowance erc-20 (token0Native=false,token1Native=false,token0Type=false,
    token1Type=true,token0NotEnoughAllowance=true,token1NotEnoughAllowance=false)
    PASS Invalid deposit - not enough allowance erc-20 (token0Native=false,token1Native=false,token0Type=true,
    token1Type=true,token0NotEnoughAllowance=true,token1NotEnoughAllowance=false)
    PASS Invalid deposit - not enough allowance erc-20 (token0Native=false,token1Native=false,token0Type=false,
    token1Type=false,token0NotEnoughAllowance=false,token1NotEnoughAllowance=true)
    PASS Invalid deposit - not enough allowance erc-20 (token0Native=true,token1Native=false,token0Type=false,
    token1Type=false,token0NotEnoughAllowance=false,token1NotEnoughAllowance=true)
    PASS Invalid deposit - not enough allowance erc-20 (token0Native=false,token1Native=false,token0Type=true,
    token1Type=false,token0NotEnoughAllowance=false,token1NotEnoughAllowance=true)
    PASS Invalid deposit - not enough allowance erc-20 (token0Native=false,token1Native=false,token0Type=false,
    token1Type=true,token0NotEnoughAllowance=false,token1NotEnoughAllowance=true)
    PASS Invalid deposit - not enough allowance erc-20 (token0Native=true,token1Native=false,token0Type=false,
    token1Type=true,token0NotEnoughAllowance=false,token1NotEnoughAllowance=true)
    PASS Invalid deposit - not enough allowance erc-20 (token0Native=false,token1Native=false,token0Type=true,
    token1Type=true,token0NotEnoughAllowance=false,token1NotEnoughAllowance=true)
    PASS Invalid deposit - not enough allowance erc-20 (token0Native=false,token1Native=false,token0Type=false,
    token1Type=false,token0NotEnoughAllowance=true,token1NotEnoughAllowance=true)
    PASS Invalid deposit - not enough allowance erc-20 (token0Native=false,token1Native=false,token0Type=true,
    token1Type=false,token0NotEnoughAllowance=true,token1NotEnoughAllowance=true)
    PASS Invalid deposit - not enough allowance erc-20 (token0Native=false,token1Native=false,token0Type=false,
    token1Type=true,token0NotEnoughAllowance=true,token1NotEnoughAllowance=true)
    PASS Invalid deposit - not enough allowance erc-20 (token0Native=false,token1Native=false,token0Type=true,
    token1Type=true,token0NotEnoughAllowance=true,token1NotEnoughAllowance=true)
    PASS Invalid deposit - not enough msg.value (token0Native=true,token1Native=false,token0Type=false,
    token1Type=false,token0NotEnoughMsgValue=true,token1NotEnoughMsgValue=false)
    PASS Invalid deposit - not enough msg.value (token0Native=true,token1Native=true,token0Type=false,
    token1Type=false,token0NotEnoughMsgValue=true,token1NotEnoughMsgValue=false)
    PASS Invalid deposit - not enough msg.value (token0Native=true,token1Native=false,token0Type=false,
    token1Type=true,token0NotEnoughMsgValue=true,token1NotEnoughMsgValue=false)
    PASS Invalid deposit - not enough msg.value (token0Native=false,token1Native=true,token0Type=false,
    token1Type=false,token0NotEnoughMsgValue=false,token1NotEnoughMsgValue=true)
    PASS Invalid deposit - not enough msg.value (token0Native=true,token1Native=true,token0Type=false,
    token1Type=false,token0NotEnoughMsgValue=false,token1NotEnoughMsgValue=true)
    PASS Invalid deposit - not enough msg.value (token0Native=false,token1Native=true,token0Type=true,
    token1Type=false,token0NotEnoughMsgValue=false,token1NotEnoughMsgValue=true)
    PASS Invalid deposit - not enough msg.value (token0Native=true,token1Native=true,token0Type=false,
    token1Type=false,token0NotEnoughMsgValue=true,token1NotEnoughMsgValue=true)
    PASS Invalid deposit - msg.value only enough for one deposit
    PASS Invalid deposit - vault not opened
    PASS Invalid deposit - vault got closed
  Front
    PASS Valid front (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
    false,noToken0=false,noToken1=false,usesExecution=false)
    PASS Valid front (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
    false,noToken0=false,noToken1=false,usesExecution=false)
    PASS Valid front (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
    false,noToken0=false,noToken1=false,usesExecution=false)
    PASS Valid front (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
    false,noToken0=false,noToken1=false,usesExecution=false)
    PASS Valid front (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=
    false,noToken0=false,noToken1=false,usesExecution=false)
```

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694       PASS Valid front (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=
          false,noToken0=false,noToken1=false,usesExecution=false)
695       PASS Valid front (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
          false,noToken0=false,noToken1=false,usesExecution=false)
696       PASS Valid front (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
          false,noToken0=false,noToken1=false,usesExecution=false)
697       PASS Valid front (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=
          false,noToken0=false,noToken1=false,usesExecution=false)
698       PASS Valid front (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
          true,noToken0=false,noToken1=false,usesExecution=false)
699       PASS Valid front (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
          true,noToken0=false,noToken1=false,usesExecution=false)
700       PASS Valid front (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
          true,noToken0=false,noToken1=false,usesExecution=false)
701       PASS Valid front (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
          true,noToken0=false,noToken1=false,usesExecution=false)
702       PASS Valid front (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=
          true,noToken0=false,noToken1=false,usesExecution=false)
703       PASS Valid front (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=
          true,noToken0=false,noToken1=false,usesExecution=false)
704       PASS Valid front (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
          true,noToken0=false,noToken1=false,usesExecution=false)
705       PASS Valid front (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
          true,noToken0=false,noToken1=false,usesExecution=false)
706       PASS Valid front (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=
          true,noToken0=false,noToken1=false,usesExecution=false)
707       PASS Valid front (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
          false,noToken0=true,noToken1=false,usesExecution=false)
708       PASS Valid front (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
          false,noToken0=true,noToken1=false,usesExecution=false)
709       PASS Valid front (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
          false,noToken0=true,noToken1=false,usesExecution=false)
710       PASS Valid front (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
          false,noToken0=true,noToken1=false,usesExecution=false)
711       PASS Valid front (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=
          false,noToken0=true,noToken1=false,usesExecution=false)
712       PASS Valid front (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=
          false,noToken0=true,noToken1=false,usesExecution=false)
713       PASS Valid front (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
          false,noToken0=true,noToken1=false,usesExecution=false)
714       PASS Valid front (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
          false,noToken0=true,noToken1=false,usesExecution=false)
715       PASS Valid front (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=
          false,noToken0=true,noToken1=false,usesExecution=false)
716       PASS Valid front (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
          true,noToken0=true,noToken1=false,usesExecution=false)
717       PASS Valid front (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
          true,noToken0=true,noToken1=false,usesExecution=false)
718       PASS Valid front (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
          true,noToken0=true,noToken1=false,usesExecution=false)
719       PASS Valid front (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
          true,noToken0=true,noToken1=false,usesExecution=false)
720       PASS Valid front (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=
          true,noToken0=true,noToken1=false,usesExecution=false)
721       PASS Valid front (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=
          true,noToken0=true,noToken1=false,usesExecution=false)
722       PASS Valid front (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
          true,noToken0=true,noToken1=false,usesExecution=false)
723       PASS Valid front (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
          true,noToken0=true,noToken1=false,usesExecution=false)
724       PASS Valid front (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=
          true,noToken0=true,noToken1=false,usesExecution=false)
725       PASS Valid front (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
          false,noToken0=false,noToken1=true,usesExecution=false)
726       PASS Valid front (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
          false,noToken0=false,noToken1=true,usesExecution=false)
727       PASS Valid front (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
          false,noToken0=false,noToken1=true,usesExecution=false)
728       PASS Valid front (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
          false,noToken0=false,noToken1=true,usesExecution=false)
```

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729     PASS Valid front (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=
        false,noToken0=false,noToken1=true,usesExecution=false)
730     PASS Valid front (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=
        false,noToken0=false,noToken1=true,usesExecution=false)
731     PASS Valid front (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
        false,noToken0=false,noToken1=true,usesExecution=false)
732     PASS Valid front (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
        false,noToken0=false,noToken1=true,usesExecution=false)
733     PASS Valid front (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=
        false,noToken0=false,noToken1=true,usesExecution=false)
734     PASS Valid front (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
        true,noToken0=false,noToken1=true,usesExecution=false)
735     PASS Valid front (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
        true,noToken0=false,noToken1=true,usesExecution=false)
736     PASS Valid front (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
        true,noToken0=false,noToken1=true,usesExecution=false)
737     PASS Valid front (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
        true,noToken0=false,noToken1=true,usesExecution=false)
738     PASS Valid front (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=
        true,noToken0=false,noToken1=true,usesExecution=false)
739     PASS Valid front (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=
        true,noToken0=false,noToken1=true,usesExecution=false)
740     PASS Valid front (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
        true,noToken0=false,noToken1=true,usesExecution=false)
741     PASS Valid front (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
        true,noToken0=false,noToken1=true,usesExecution=false)
742     PASS Valid front (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=
        true,noToken0=false,noToken1=true,usesExecution=false)
743     PASS Valid front (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
        false,noToken0=false,noToken1=false,usesExecution=true)
744     PASS Valid front (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
        false,noToken0=false,noToken1=false,usesExecution=true)
745     PASS Valid front (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
        false,noToken0=false,noToken1=false,usesExecution=true)
746     PASS Valid front (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
        false,noToken0=false,noToken1=false,usesExecution=true)
747     PASS Valid front (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=
        false,noToken0=false,noToken1=false,usesExecution=true)
748     PASS Valid front (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=
        false,noToken0=false,noToken1=false,usesExecution=true)
749     PASS Valid front (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
        false,noToken0=false,noToken1=false,usesExecution=true)
750     PASS Valid front (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
        false,noToken0=false,noToken1=false,usesExecution=true)
751     PASS Valid front (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=
        false,noToken0=false,noToken1=false,usesExecution=true)
752     PASS Valid front (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
        true,noToken0=false,noToken1=false,usesExecution=true)
753     PASS Valid front (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
        true,noToken0=false,noToken1=false,usesExecution=true)
754     PASS Valid front (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
        true,noToken0=false,noToken1=false,usesExecution=true)
755     PASS Valid front (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
        true,noToken0=false,noToken1=false,usesExecution=true)
756     PASS Valid front (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=
        true,noToken0=false,noToken1=false,usesExecution=true)
757     PASS Valid front (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=
        true,noToken0=false,noToken1=false,usesExecution=true)
758     PASS Valid front (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
        true,noToken0=false,noToken1=false,usesExecution=true)
759     PASS Valid front (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
        true,noToken0=false,noToken1=false,usesExecution=true)
760     PASS Valid front (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=
        true,noToken0=false,noToken1=false,usesExecution=true)
761     PASS Valid front (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
        false,noToken0=true,noToken1=false,usesExecution=true)
762     PASS Valid front (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
        false,noToken0=true,noToken1=false,usesExecution=true)
763     PASS Valid front (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
        false,noToken0=true,noToken1=false,usesExecution=true)
```

```
764   PASS Valid front (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
      false,noToken0=true,noToken1=false,usesExecution=true)
765   PASS Valid front (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=
      false,noToken0=true,noToken1=false,usesExecution=true)
766   PASS Valid front (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=
      false,noToken0=true,noToken1=false,usesExecution=true)
767   PASS Valid front (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
      false,noToken0=true,noToken1=false,usesExecution=true)
768   PASS Valid front (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
      false,noToken0=true,noToken1=false,usesExecution=true)
769   PASS Valid front (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=
      false,noToken0=true,noToken1=false,usesExecution=true)
770   PASS Valid front (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=true)
771   PASS Valid front (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=true)
772   PASS Valid front (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=true)
773   PASS Valid front (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=true)
774   PASS Valid front (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=true)
775   PASS Valid front (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=true)
776   PASS Valid front (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=true)
777   PASS Valid front (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=true)
778   PASS Valid front (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=true)
779   PASS Valid front (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=true)
780   PASS Valid front (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=true)
781   PASS Valid front (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=true)
782   PASS Valid front (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=true)
783   PASS Valid front (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=true)
784   PASS Valid front (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=true)
785   PASS Valid front (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=true)
786   PASS Valid front (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=true)
787   PASS Valid front (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=true)
788   PASS Valid front (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
      true,noToken0=false,noToken1=true,usesExecution=true)
789   PASS Valid front (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
      true,noToken0=false,noToken1=true,usesExecution=true)
790   PASS Valid front (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
      true,noToken0=false,noToken1=true,usesExecution=true)
791   PASS Valid front (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
      true,noToken0=false,noToken1=true,usesExecution=true)
792   PASS Valid front (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=
      true,noToken0=false,noToken1=true,usesExecution=true)
793   PASS Valid front (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=
      true,noToken0=false,noToken1=true,usesExecution=true)
794   PASS Valid front (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
      true,noToken0=false,noToken1=true,usesExecution=true)
795   PASS Valid front (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
      true,noToken0=false,noToken1=true,usesExecution=true)
796   PASS Valid front (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=
      true,noToken0=false,noToken1=true,usesExecution=true)
797   PASS Invalid front - not enough balance erc-20 (token0Native=false,token1Native=false,token0Type=false,
      token1Type=false,token0NotEnoughBalance=true,token1NotEnoughBalance=false)
798   PASS Invalid front - not enough balance erc-20 (token0Native=false,token1Native=true,token0Type=false,
      token1Type=false,token0NotEnoughBalance=true,token1NotEnoughBalance=false)
```

```
799   PASS Invalid front - not enough balance erc-20 (token0Native=false,token1Native=false,token0Type=true,
      token1Type=false,token0NotEnoughBalance=true,token1NotEnoughBalance=false)
800   PASS Invalid front - not enough balance erc-20 (token0Native=false,token1Native=true,token0Type=true,
      token1Type=false,token0NotEnoughBalance=true,token1NotEnoughBalance=false)
801   PASS Invalid front - not enough balance erc-20 (token0Native=false,token1Native=false,token0Type=false,
      token1Type=true,token0NotEnoughBalance=true,token1NotEnoughBalance=false)
802   PASS Invalid front - not enough balance erc-20 (token0Native=false,token1Native=false,token0Type=true,
      token1Type=true,token0NotEnoughBalance=true,token1NotEnoughBalance=false)
803   PASS Invalid front - not enough balance erc-20 (token0Native=false,token1Native=false,token0Type=false,
      token1Type=false,token0NotEnoughBalance=false,token1NotEnoughBalance=true)
804   PASS Invalid front - not enough balance erc-20 (token0Native=true,token1Native=false,token0Type=false,
      token1Type=false,token0NotEnoughBalance=false,token1NotEnoughBalance=true)
805   PASS Invalid front - not enough balance erc-20 (token0Native=false,token1Native=false,token0Type=true,
      token1Type=false,token0NotEnoughBalance=false,token1NotEnoughBalance=true)
806   PASS Invalid front - not enough balance erc-20 (token0Native=false,token1Native=false,token0Type=false,
      token1Type=true,token0NotEnoughBalance=false,token1NotEnoughBalance=true)
807   PASS Invalid front - not enough balance erc-20 (token0Native=true,token1Native=false,token0Type=false,
      token1Type=true,token0NotEnoughBalance=false,token1NotEnoughBalance=true)
808   PASS Invalid front - not enough balance erc-20 (token0Native=false,token1Native=false,token0Type=true,
      token1Type=true,token0NotEnoughBalance=false,token1NotEnoughBalance=true)
809   PASS Invalid front - not enough balance erc-20 (token0Native=false,token1Native=false,token0Type=false,
      token1Type=false,token0NotEnoughBalance=true,token1NotEnoughBalance=true)
810   PASS Invalid front - not enough balance erc-20 (token0Native=false,token1Native=false,token0Type=true,
      token1Type=false,token0NotEnoughBalance=true,token1NotEnoughBalance=true)
811   PASS Invalid front - not enough balance erc-20 (token0Native=false,token1Native=false,token0Type=false,
      token1Type=true,token0NotEnoughBalance=true,token1NotEnoughBalance=true)
812   PASS Invalid front - not enough balance erc-20 (token0Native=false,token1Native=false,token0Type=true,
      token1Type=true,token0NotEnoughBalance=true,token1NotEnoughBalance=true)
813   PASS Invalid front - not enough allowance erc-20 (token0Native=false,token1Native=false,token0Type=false,
      token1Type=false,token0NotEnoughAllowance=true,token1NotEnoughAllowance=false)
814   PASS Invalid front - not enough allowance erc-20 (token0Native=false,token1Native=true,token0Type=false,
      token1Type=false,token0NotEnoughAllowance=true,token1NotEnoughAllowance=false)
815   PASS Invalid front - not enough allowance erc-20 (token0Native=false,token1Native=false,token0Type=true,
      token1Type=false,token0NotEnoughAllowance=true,token1NotEnoughAllowance=false)
816   PASS Invalid front - not enough allowance erc-20 (token0Native=false,token1Native=true,token0Type=true,
      token1Type=false,token0NotEnoughAllowance=true,token1NotEnoughAllowance=false)
817   PASS Invalid front - not enough allowance erc-20 (token0Native=false,token1Native=false,token0Type=false,
      token1Type=true,token0NotEnoughAllowance=true,token1NotEnoughAllowance=false)
818   PASS Invalid front - not enough allowance erc-20 (token0Native=false,token1Native=false,token0Type=true,
      token1Type=true,token0NotEnoughAllowance=true,token1NotEnoughAllowance=false)
819   PASS Invalid front - not enough allowance erc-20 (token0Native=false,token1Native=false,token0Type=false,
      token1Type=false,token0NotEnoughAllowance=false,token1NotEnoughAllowance=true)
820   PASS Invalid front - not enough allowance erc-20 (token0Native=true,token1Native=false,token0Type=false,
      token1Type=false,token0NotEnoughAllowance=false,token1NotEnoughAllowance=true)
821   PASS Invalid front - not enough allowance erc-20 (token0Native=false,token1Native=false,token0Type=true,
      token1Type=false,token0NotEnoughAllowance=false,token1NotEnoughAllowance=true)
822   PASS Invalid front - not enough allowance erc-20 (token0Native=false,token1Native=false,token0Type=false,
      token1Type=true,token0NotEnoughAllowance=false,token1NotEnoughAllowance=true)
823   PASS Invalid front - not enough allowance erc-20 (token0Native=true,token1Native=false,token0Type=false,
      token1Type=true,token0NotEnoughAllowance=false,token1NotEnoughAllowance=true)
824   PASS Invalid front - not enough allowance erc-20 (token0Native=false,token1Native=false,token0Type=true,
      token1Type=true,token0NotEnoughAllowance=false,token1NotEnoughAllowance=true)
825   PASS Invalid front - not enough allowance erc-20 (token0Native=false,token1Native=false,token0Type=false,
      token1Type=false,token0NotEnoughAllowance=true,token1NotEnoughAllowance=true)
826   PASS Invalid front - not enough allowance erc-20 (token0Native=false,token1Native=false,token0Type=true,
      token1Type=false,token0NotEnoughAllowance=true,token1NotEnoughAllowance=true)
827   PASS Invalid front - not enough allowance erc-20 (token0Native=false,token1Native=false,token0Type=false,
      token1Type=true,token0NotEnoughAllowance=true,token1NotEnoughAllowance=true)
828   PASS Invalid front - not enough allowance erc-20 (token0Native=false,token1Native=false,token0Type=true,
      token1Type=true,token0NotEnoughAllowance=true,token1NotEnoughAllowance=true)
829   PASS Invalid front - not enough msg.value (token0Native=true,token1Native=false,token0Type=false,token1Type
      =false,token0NotEnoughMsgValue=true,token1NotEnoughMsgValue=false)
830   PASS Invalid front - not enough msg.value (token0Native=true,token1Native=true,token0Type=false,token1Type=
      false,token0NotEnoughMsgValue=true,token1NotEnoughMsgValue=false)
831   PASS Invalid front - not enough msg.value (token0Native=true,token1Native=false,token0Type=false,token1Type
      =true,token0NotEnoughMsgValue=true,token1NotEnoughMsgValue=false)
832   PASS Invalid front - not enough msg.value (token0Native=false,token1Native=true,token0Type=false,token1Type
      =false,token0NotEnoughMsgValue=false,token1NotEnoughMsgValue=true)
833   PASS Invalid front - not enough msg.value (token0Native=true,token1Native=true,token0Type=false,token1Type=
      false,token0NotEnoughMsgValue=false,token1NotEnoughMsgValue=true)
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```
      PASS Invalid front - not enough msg.value (token0Native=false,token1Native=true,token0Type=true,token1Type=
    false,token0NotEnoughMsgValue=false,token1NotEnoughMsgValue=true)
      PASS Invalid front - not enough msg.value (token0Native=true,token1Native=true,token0Type=false,token1Type=
    false,token0NotEnoughMsgValue=true,token1NotEnoughMsgValue=true)
      PASS Invalid front - msg.value only enough for one deposit
      PASS Invalid front - vault not opened
      PASS Invalid front - vault got closed
      PASS Invalid front - already fronted
      PASS Invalid front - already claimed
      PASS Invalid front - amount0 overflow 64-bits
    Claim
      PASS Valid claim (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
    false,noToken0=false,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
    false,noToken0=false,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
    false,noToken0=false,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
    false,noToken0=false,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=
    false,noToken0=false,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=
    false,noToken0=false,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
    false,noToken0=false,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
    false,noToken0=false,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=
    false,noToken0=false,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
    true,noToken0=false,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
    true,noToken0=false,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
    true,noToken0=false,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
    true,noToken0=false,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=
    true,noToken0=false,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=
    true,noToken0=false,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
    true,noToken0=false,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
    true,noToken0=false,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=
    true,noToken0=false,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
    false,noToken0=true,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
    false,noToken0=true,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
    false,noToken0=true,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
    false,noToken0=true,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=
    false,noToken0=true,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=
    false,noToken0=true,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
    false,noToken0=true,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
    false,noToken0=true,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=
    false,noToken0=true,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
    true,noToken0=true,noToken1=false,usesExecution=false)
      PASS Valid claim (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
    true,noToken0=true,noToken1=false,usesExecution=false)
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872       PASS Valid claim (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=false)
873       PASS Valid claim (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=false)
874       PASS Valid claim (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=false) (51ms)
875       PASS Valid claim (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=false)
876       PASS Valid claim (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=false)
877       PASS Valid claim (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=false)
878       PASS Valid claim (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=false)
879       PASS Valid claim (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=false)
880       PASS Valid claim (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=false)
881       PASS Valid claim (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=false)
882       PASS Valid claim (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=false)
883       PASS Valid claim (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=false)
884       PASS Valid claim (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=false)
885       PASS Valid claim (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=false)
886       PASS Valid claim (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=false)
887       PASS Valid claim (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=false)
888       PASS Valid claim (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
      true,noToken0=false,noToken1=true,usesExecution=false)
889       PASS Valid claim (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
      true,noToken0=false,noToken1=true,usesExecution=false)
890       PASS Valid claim (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
      true,noToken0=false,noToken1=true,usesExecution=false)
891       PASS Valid claim (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
      true,noToken0=false,noToken1=true,usesExecution=false)
892       PASS Valid claim (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=
      true,noToken0=false,noToken1=true,usesExecution=false)
893       PASS Valid claim (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=
      true,noToken0=false,noToken1=true,usesExecution=false)
894       PASS Valid claim (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
      true,noToken0=false,noToken1=true,usesExecution=false)
895       PASS Valid claim (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
      true,noToken0=false,noToken1=true,usesExecution=false)
896       PASS Valid claim (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=
      true,noToken0=false,noToken1=true,usesExecution=false)
897       PASS Valid claim (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
      false,noToken0=false,noToken1=false,usesExecution=true)
898       PASS Valid claim (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
      false,noToken0=false,noToken1=false,usesExecution=true)
899       PASS Valid claim (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
      false,noToken0=false,noToken1=false,usesExecution=true)
900       PASS Valid claim (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
      false,noToken0=false,noToken1=false,usesExecution=true)
901       PASS Valid claim (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=
      false,noToken0=false,noToken1=false,usesExecution=true)
902       PASS Valid claim (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=
      false,noToken0=false,noToken1=false,usesExecution=true) (47ms)
903       PASS Valid claim (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
      false,noToken0=false,noToken1=false,usesExecution=true) (65ms)
904       PASS Valid claim (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
      false,noToken0=false,noToken1=false,usesExecution=true)
905       PASS Valid claim (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=
      false,noToken0=false,noToken1=false,usesExecution=true) (41ms)
906       PASS Valid claim (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
      true,noToken0=false,noToken1=false,usesExecution=true) (39ms)
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907       PASS Valid claim (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
      true,noToken0=false,noToken1=false,usesExecution=true)
908       PASS Valid claim (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
      true,noToken0=false,noToken1=false,usesExecution=true) (45ms)
909       PASS Valid claim (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
      true,noToken0=false,noToken1=false,usesExecution=true)
910       PASS Valid claim (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=
      true,noToken0=false,noToken1=false,usesExecution=true) (39ms)
911       PASS Valid claim (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=
      true,noToken0=false,noToken1=false,usesExecution=true)
912       PASS Valid claim (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
      true,noToken0=false,noToken1=false,usesExecution=true) (41ms)
913       PASS Valid claim (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
      true,noToken0=false,noToken1=false,usesExecution=true) (39ms)
914       PASS Valid claim (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=
      true,noToken0=false,noToken1=false,usesExecution=true) (40ms)
915       PASS Valid claim (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
      false,noToken0=true,noToken1=false,usesExecution=true) (41ms)
916       PASS Valid claim (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
      false,noToken0=true,noToken1=false,usesExecution=true)
917       PASS Valid claim (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
      false,noToken0=true,noToken1=false,usesExecution=true) (40ms)
918       PASS Valid claim (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
      false,noToken0=true,noToken1=false,usesExecution=true) (38ms)
919       PASS Valid claim (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=
      false,noToken0=true,noToken1=false,usesExecution=true) (40ms)
920       PASS Valid claim (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=
      false,noToken0=true,noToken1=false,usesExecution=true)
921       PASS Valid claim (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
      false,noToken0=true,noToken1=false,usesExecution=true) (42ms)
922       PASS Valid claim (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
      false,noToken0=true,noToken1=false,usesExecution=true) (42ms)
923       PASS Valid claim (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=
      false,noToken0=true,noToken1=false,usesExecution=true) (57ms)
924       PASS Valid claim (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=true) (49ms)
925       PASS Valid claim (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=true) (48ms)
926       PASS Valid claim (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=true) (67ms)
927       PASS Valid claim (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=true) (56ms)
928       PASS Valid claim (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=true) (48ms)
929       PASS Valid claim (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=true)
930       PASS Valid claim (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=true)
931       PASS Valid claim (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=true)
932       PASS Valid claim (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=
      true,noToken0=true,noToken1=false,usesExecution=true)
933       PASS Valid claim (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=true) (39ms)
934       PASS Valid claim (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=true)
935       PASS Valid claim (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=true)
936       PASS Valid claim (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=true)
937       PASS Valid claim (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=true)
938       PASS Valid claim (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=true)
939       PASS Valid claim (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=true) (41ms)
940       PASS Valid claim (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=true) (57ms)
941       PASS Valid claim (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=
      false,noToken0=false,noToken1=true,usesExecution=true)
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  PASS Valid claim (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=true)
  PASS Valid claim (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=true)
  PASS Valid claim (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=true)
  PASS Valid claim (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=true)
  PASS Valid claim (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=true)
  PASS Valid claim (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=true)
  PASS Valid claim (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=true)
  PASS Valid claim (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=true)
  PASS Valid claim (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=true)
  PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=false,noToken0=false,noToken1=false,usesExecution=false)
  PASS Valid claim - fronted (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=false,noToken0=false,noToken1=false,usesExecution=false)
  PASS Valid claim - fronted (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=false,noToken0=false,noToken1=false,usesExecution=false)
  PASS Valid claim - fronted (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=false,noToken0=false,noToken1=false,usesExecution=false)
  PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=false,noToken0=false,noToken1=false,usesExecution=false) (44ms)
  PASS Valid claim - fronted (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=false,noToken0=false,noToken1=false,usesExecution=false)
  PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=false,noToken0=false,noToken1=false,usesExecution=false)
  PASS Valid claim - fronted (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=false,noToken0=false,noToken1=false,usesExecution=false)
  PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=false,noToken0=false,noToken1=false,usesExecution=false)
  PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=true,noToken0=false,noToken1=false,usesExecution=false)
  PASS Valid claim - fronted (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=true,noToken0=false,noToken1=false,usesExecution=false) (44ms)
  PASS Valid claim - fronted (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=true,noToken0=false,noToken1=false,usesExecution=false)
  PASS Valid claim - fronted (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=true,noToken0=false,noToken1=false,usesExecution=false)
  PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=true,noToken0=false,noToken1=false,usesExecution=false)
  PASS Valid claim - fronted (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=true,noToken0=false,noToken1=false,usesExecution=false)
  PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=true,noToken0=false,noToken1=false,usesExecution=false)
  PASS Valid claim - fronted (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=true,noToken0=false,noToken1=false,usesExecution=false)
  PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=true,token1Type=true,thirdPartyFront=true,noToken0=false,noToken1=false,usesExecution=false) (40ms)
  PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=false,noToken0=true,noToken1=false,usesExecution=false)
  PASS Valid claim - fronted (token0Native=true,token1Native=false,token0Type=false,token1Type=false,thirdPartyFront=false,noToken0=true,noToken1=false,usesExecution=false)
  PASS Valid claim - fronted (token0Native=false,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=false,noToken0=true,noToken1=false,usesExecution=false)
  PASS Valid claim - fronted (token0Native=true,token1Native=true,token0Type=false,token1Type=false,thirdPartyFront=false,noToken0=true,noToken1=false,usesExecution=false)
  PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=true,token1Type=false,thirdPartyFront=false,noToken0=true,noToken1=false,usesExecution=false)
  PASS Valid claim - fronted (token0Native=false,token1Native=true,token0Type=true,token1Type=false,thirdPartyFront=false,noToken0=true,noToken1=false,usesExecution=false) (38ms)
  PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=false,noToken0=true,noToken1=false,usesExecution=false) (42ms)
  PASS Valid claim - fronted (token0Native=true,token1Native=false,token0Type=false,token1Type=true,thirdPartyFront=false,noToken0=true,noToken1=false,usesExecution=false)
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977   PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=true,token1Type=true,
      thirdPartyFront=false,noToken0=true,noToken1=false,usesExecution=false)
978   PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=false,token1Type=false,
      thirdPartyFront=true,noToken0=true,noToken1=false,usesExecution=false)
979   PASS Valid claim - fronted (token0Native=true,token1Native=false,token0Type=false,token1Type=false,
      thirdPartyFront=true,noToken0=true,noToken1=false,usesExecution=false)
980   PASS Valid claim - fronted (token0Native=false,token1Native=true,token0Type=false,token1Type=false,
      thirdPartyFront=true,noToken0=true,noToken1=false,usesExecution=false)
981   PASS Valid claim - fronted (token0Native=true,token1Native=true,token0Type=false,token1Type=false,
      thirdPartyFront=true,noToken0=true,noToken1=false,usesExecution=false)
982   PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=true,token1Type=false,
      thirdPartyFront=true,noToken0=true,noToken1=false,usesExecution=false) (43ms)
983   PASS Valid claim - fronted (token0Native=false,token1Native=true,token0Type=true,token1Type=false,
      thirdPartyFront=true,noToken0=true,noToken1=false,usesExecution=false) (38ms)
984   PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=false,token1Type=true,
      thirdPartyFront=true,noToken0=true,noToken1=false,usesExecution=false)
985   PASS Valid claim - fronted (token0Native=true,token1Native=false,token0Type=false,token1Type=true,
      thirdPartyFront=true,noToken0=true,noToken1=false,usesExecution=false)
986   PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=true,token1Type=true,
      thirdPartyFront=true,noToken0=true,noToken1=false,usesExecution=false)
987   PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=false,token1Type=false,
      thirdPartyFront=false,noToken0=false,noToken1=true,usesExecution=false)
988   PASS Valid claim - fronted (token0Native=true,token1Native=false,token0Type=false,token1Type=false,
      thirdPartyFront=false,noToken0=false,noToken1=true,usesExecution=false)
989   PASS Valid claim - fronted (token0Native=false,token1Native=true,token0Type=false,token1Type=false,
      thirdPartyFront=false,noToken0=false,noToken1=true,usesExecution=false)
990   PASS Valid claim - fronted (token0Native=true,token1Native=true,token0Type=false,token1Type=false,
      thirdPartyFront=false,noToken0=false,noToken1=true,usesExecution=false)
991   PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=true,token1Type=false,
      thirdPartyFront=false,noToken0=false,noToken1=true,usesExecution=false)
992   PASS Valid claim - fronted (token0Native=false,token1Native=true,token0Type=true,token1Type=false,
      thirdPartyFront=false,noToken0=false,noToken1=true,usesExecution=false) (40ms)
993   PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=false,token1Type=true,
      thirdPartyFront=false,noToken0=false,noToken1=true,usesExecution=false)
994   PASS Valid claim - fronted (token0Native=true,token1Native=false,token0Type=false,token1Type=true,
      thirdPartyFront=false,noToken0=false,noToken1=true,usesExecution=false)
995   PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=true,token1Type=true,
      thirdPartyFront=false,noToken0=false,noToken1=true,usesExecution=false)
996   PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=false,token1Type=false,
      thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=false)
997   PASS Valid claim - fronted (token0Native=true,token1Native=false,token0Type=false,token1Type=false,
      thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=false)
998   PASS Valid claim - fronted (token0Native=false,token1Native=true,token0Type=false,token1Type=false,
      thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=false)
999   PASS Valid claim - fronted (token0Native=true,token1Native=true,token0Type=false,token1Type=false,
      thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=false)
1000  PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=true,token1Type=false,
      thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=false) (39ms)
1001  PASS Valid claim - fronted (token0Native=false,token1Native=true,token0Type=true,token1Type=false,
      thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=false)
1002  PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=false,token1Type=true,
      thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=false) (48ms)
1003  PASS Valid claim - fronted (token0Native=true,token1Native=false,token0Type=false,token1Type=true,
      thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=false)
1004  PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=true,token1Type=true,
      thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=false)
1005  PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=false,token1Type=false,
      thirdPartyFront=false,noToken0=false,noToken1=false,usesExecution=true) (40ms)
1006  PASS Valid claim - fronted (token0Native=true,token1Native=false,token0Type=false,token1Type=false,
      thirdPartyFront=false,noToken0=false,noToken1=false,usesExecution=true)
1007  PASS Valid claim - fronted (token0Native=false,token1Native=true,token0Type=false,token1Type=false,
      thirdPartyFront=false,noToken0=false,noToken1=false,usesExecution=true)
1008  PASS Valid claim - fronted (token0Native=true,token1Native=true,token0Type=false,token1Type=false,
      thirdPartyFront=false,noToken0=false,noToken1=false,usesExecution=true)
1009  PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=true,token1Type=false,
      thirdPartyFront=false,noToken0=false,noToken1=false,usesExecution=true)
1010  PASS Valid claim - fronted (token0Native=false,token1Native=true,token0Type=true,token1Type=false,
      thirdPartyFront=false,noToken0=false,noToken1=false,usesExecution=true)
1011  PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=false,token1Type=true,
      thirdPartyFront=false,noToken0=false,noToken1=false,usesExecution=true) (40ms)
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1012      PASS Valid claim - fronted (token0Native=true,token1Native=false,token0Type=false,token1Type=true,
          thirdPartyFront=false,noToken0=false,noToken1=false,usesExecution=true)
1013      PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=true,token1Type=true,
          thirdPartyFront=false,noToken0=false,noToken1=false,usesExecution=true)
1014      PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=false,token1Type=false,
          thirdPartyFront=true,noToken0=false,noToken1=false,usesExecution=true)
1015      PASS Valid claim - fronted (token0Native=true,token1Native=false,token0Type=false,token1Type=false,
          thirdPartyFront=true,noToken0=false,noToken1=false,usesExecution=true)
1016      PASS Valid claim - fronted (token0Native=false,token1Native=true,token0Type=false,token1Type=false,
          thirdPartyFront=true,noToken0=false,noToken1=false,usesExecution=true) (38ms)
1017      PASS Valid claim - fronted (token0Native=true,token1Native=true,token0Type=false,token1Type=false,
          thirdPartyFront=true,noToken0=false,noToken1=false,usesExecution=true)
1018      PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=true,token1Type=false,
          thirdPartyFront=true,noToken0=false,noToken1=false,usesExecution=true) (44ms)
1019      PASS Valid claim - fronted (token0Native=false,token1Native=true,token0Type=true,token1Type=false,
          thirdPartyFront=true,noToken0=false,noToken1=false,usesExecution=true) (42ms)
1020      PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=false,token1Type=true,
          thirdPartyFront=true,noToken0=false,noToken1=false,usesExecution=true) (46ms)
1021      PASS Valid claim - fronted (token0Native=true,token1Native=false,token0Type=false,token1Type=true,
          thirdPartyFront=true,noToken0=false,noToken1=false,usesExecution=true) (42ms)
1022      PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=true,token1Type=true,
          thirdPartyFront=true,noToken0=false,noToken1=false,usesExecution=true)
1023      PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=false,token1Type=false,
          thirdPartyFront=false,noToken0=true,noToken1=false,usesExecution=true)
1024      PASS Valid claim - fronted (token0Native=true,token1Native=false,token0Type=false,token1Type=false,
          thirdPartyFront=false,noToken0=true,noToken1=false,usesExecution=true)
1025      PASS Valid claim - fronted (token0Native=false,token1Native=true,token0Type=false,token1Type=false,
          thirdPartyFront=false,noToken0=true,noToken1=false,usesExecution=true) (38ms)
1026      PASS Valid claim - fronted (token0Native=true,token1Native=true,token0Type=false,token1Type=false,
          thirdPartyFront=false,noToken0=true,noToken1=false,usesExecution=true)
1027      PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=true,token1Type=false,
          thirdPartyFront=false,noToken0=true,noToken1=false,usesExecution=true) (44ms)
1028      PASS Valid claim - fronted (token0Native=false,token1Native=true,token0Type=true,token1Type=false,
          thirdPartyFront=false,noToken0=true,noToken1=false,usesExecution=true)
1029      PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=false,token1Type=true,
          thirdPartyFront=false,noToken0=true,noToken1=false,usesExecution=true) (40ms)
1030      PASS Valid claim - fronted (token0Native=true,token1Native=false,token0Type=false,token1Type=true,
          thirdPartyFront=false,noToken0=true,noToken1=false,usesExecution=true) (42ms)
1031      PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=true,token1Type=true,
          thirdPartyFront=false,noToken0=true,noToken1=false,usesExecution=true)
1032      PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=false,token1Type=false,
          thirdPartyFront=true,noToken0=true,noToken1=false,usesExecution=true) (44ms)
1033      PASS Valid claim - fronted (token0Native=true,token1Native=false,token0Type=false,token1Type=false,
          thirdPartyFront=true,noToken0=true,noToken1=false,usesExecution=true) (44ms)
1034      PASS Valid claim - fronted (token0Native=false,token1Native=true,token0Type=false,token1Type=false,
          thirdPartyFront=true,noToken0=true,noToken1=false,usesExecution=true) (44ms)
1035      PASS Valid claim - fronted (token0Native=true,token1Native=true,token0Type=false,token1Type=false,
          thirdPartyFront=true,noToken0=true,noToken1=false,usesExecution=true) (40ms)
1036      PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=true,token1Type=false,
          thirdPartyFront=true,noToken0=true,noToken1=false,usesExecution=true) (48ms)
1037      PASS Valid claim - fronted (token0Native=false,token1Native=true,token0Type=true,token1Type=false,
          thirdPartyFront=true,noToken0=true,noToken1=false,usesExecution=true) (44ms)
1038      PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=false,token1Type=true,
          thirdPartyFront=true,noToken0=true,noToken1=false,usesExecution=true) (44ms)
1039      PASS Valid claim - fronted (token0Native=true,token1Native=false,token0Type=false,token1Type=true,
          thirdPartyFront=true,noToken0=true,noToken1=false,usesExecution=true)
1040      PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=true,token1Type=true,
          thirdPartyFront=true,noToken0=true,noToken1=false,usesExecution=true) (40ms)
1041      PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=false,token1Type=false,
          thirdPartyFront=false,noToken0=false,noToken1=true,usesExecution=true)
1042      PASS Valid claim - fronted (token0Native=true,token1Native=false,token0Type=false,token1Type=false,
          thirdPartyFront=false,noToken0=false,noToken1=true,usesExecution=true) (38ms)
1043      PASS Valid claim - fronted (token0Native=false,token1Native=true,token0Type=false,token1Type=false,
          thirdPartyFront=false,noToken0=false,noToken1=true,usesExecution=true)
1044      PASS Valid claim - fronted (token0Native=true,token1Native=true,token0Type=false,token1Type=false,
          thirdPartyFront=false,noToken0=false,noToken1=true,usesExecution=true)
1045      PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=true,token1Type=false,
          thirdPartyFront=false,noToken0=false,noToken1=true,usesExecution=true) (42ms)
1046      PASS Valid claim - fronted (token0Native=false,token1Native=true,token0Type=true,token1Type=false,
          thirdPartyFront=false,noToken0=false,noToken1=true,usesExecution=true) (47ms)
```

```
      PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=false,token1Type=true,
    thirdPartyFront=false,noToken0=false,noToken1=true,usesExecution=true) (43ms)
      PASS Valid claim - fronted (token0Native=true,token1Native=false,token0Type=false,token1Type=true,
    thirdPartyFront=false,noToken0=false,noToken1=true,usesExecution=true) (51ms)
      PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=true,token1Type=true,
    thirdPartyFront=false,noToken0=false,noToken1=true,usesExecution=true) (41ms)
      PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=false,token1Type=false,
    thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=true) (39ms)
      PASS Valid claim - fronted (token0Native=true,token1Native=false,token0Type=false,token1Type=false,
    thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=true) (50ms)
      PASS Valid claim - fronted (token0Native=false,token1Native=true,token0Type=false,token1Type=false,
    thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=true) (38ms)
      PASS Valid claim - fronted (token0Native=true,token1Native=true,token0Type=false,token1Type=false,
    thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=true)
      PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=true,token1Type=false,
    thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=true) (46ms)
      PASS Valid claim - fronted (token0Native=false,token1Native=true,token0Type=true,token1Type=false,
    thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=true)
      PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=false,token1Type=true,
    thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=true) (44ms)
      PASS Valid claim - fronted (token0Native=true,token1Native=false,token0Type=false,token1Type=true,
    thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=true) (40ms)
      PASS Valid claim - fronted (token0Native=false,token1Native=false,token0Type=true,token1Type=true,
    thirdPartyFront=true,noToken0=false,noToken1=true,usesExecution=true)
      PASS Invalid claim - vault not opened
      PASS Invalid claim - invalid tx confirmations
      PASS Invalid claim - invalid merkle proof
      PASS Invalid claim - invalid merkle proof (position)
      PASS Invalid claim - btc tx empty inputs
      PASS Invalid claim - btc tx empty inputs
      PASS Invalid claim (vault close) - output 1 not found
      PASS Invalid claim (vault close) - input 1 not found
      PASS Invalid claim (vault close) - output 1 empty script
      PASS Invalid claim (vault close) - output 1 not OP_RETURN
      PASS Invalid claim (vault close) - output 1 invalid len
      PASS Invalid claim (vault close) - caller fee 0 overflow
      PASS Invalid claim (vault close) - fronting fee 0 overflow
      PASS Invalid claim (vault close) - execution fee 0 overflow
      PASS Invalid claim (vault close) - caller fee 1 overflow
      PASS Invalid claim (vault close) - fronting fee 1 overflow
      PASS Invalid claim (vault close) - amount 0 sum overflow
      PASS Invalid claim (vault close) - amount 1 sum overflow
      PASS Invalid claim (vault close) - withdraw too much token0
      PASS Invalid claim (vault close) - withdraw too much token1

  BitcoinNoncedOutputClaimHandler
    PASS Valid random witness (108ms)
    1) Valid real witness
    PASS Invalid empty witness
    PASS Invalid incorrect commitment witness
    PASS Invalid block confirmations
    PASS Invalid merkle proof, root doesn't match
    PASS Invalid blockheader, provided header is not known to the btc relay
    PASS Invalid vout of bounds
    PASS Invalid txoHash doesn't match (due to wrong outputAmount)
    PASS Invalid txoHash doesn't match (due to wrong outputScript)
    PASS Invalid txoHash doesn't match (due to wrong nonce)

  BitcoinOutputClaimHandler
    PASS Valid random witness (107ms)
    2) Valid real witness
    PASS Invalid empty witness
    PASS Invalid incorrect commitment witness
    PASS Invalid block confirmations
    PASS Invalid merkle proof, root doesn't match
    PASS Invalid blockheader, provided header is not known to the btc relay
    PASS Invalid vout of bounds
    PASS Invalid txoHash doesn't match (due to wrong outputAmount)
    PASS Invalid txoHash doesn't match (due to wrong outputScript)

```

```
CompactBlockHeader
  PASS Valid verify out of bounds
  PASS Valid verify out of bounds with offset
  PASS Invalid verify out of bounds
  PASS Valid read values
  3) Existing blockheaders

Difficulty
  PASS Get chainwork random data
  4) Compute new target real adjustments
  5) Get chainwork real data

Nbits
  PASS Nbits to target
  PASS Target to nbits
  PASS Bitcoin core test vector
  PASS Bitcoin core negative nbits test vector
  PASS Random targets

StoredBlockHeader
  PASS Valid from calldata
  PASS Valid from calldata with offset
  PASS Invalid from calldata with offset
  PASS Valid read values
  6) Existing blockheaders
  PASS Valid parse random (107ms)
  PASS Valid update random (767ms)
  PASS Valid update random block on PoW readjustment (955ms)
  PASS Valid update random block on PoW readjustment, too fast (43ms)
  PASS Valid update random block on PoW readjustment, too slow (492ms)
  PASS Valid update random block, with timestamp larger than median of last 11 blocks (706ms)
  PASS Invalid update random block, due to low PoW (4348ms)
  PASS Invalid update random block, due to wrong nBits (289ms)
  PASS Invalid update random block, due to wrong nBits during difficulty retarget (226ms)
  PASS Invalid update random block, due to wrong previous blockhash (611ms)
  PASS Invalid update random block, due to timestamp not being larger than median of last 11 blocks (786ms)
  PASS Invalid update random block, due to timestamp being too far in the future (555ms)
  7) Valid update real
  8) Valid update real on PoW readjustment
  9) Valid update real on PoW readjustment, too fast

BitcoinTxIdClaimHandler
  PASS Valid random witness (96ms)
  10) Valid real witness
  PASS Invalid empty witness
  PASS Invalid incorrect commitment witness
  PASS Invalid block confirmations
  PASS Invalid merkle proof, root doesn't match
  PASS Invalid blockheader, provided header is not known to the btc relay

BitcoinMerkleTree
  PASS Randomly generated tests
  11) Real data tests

BitcoinTx
  PASS Valid random bitcoin txs (512ms)
  12) Valid real bitcoin txs
  PASS Invalid witness not stripped
  PASS Invalid tx more than expected data
  PASS Invalid tx length exactly 64 bytes

Endianness
  PASS Reverse uint32
  PASS Reverse uint64
  PASS Reverse bytes32

EIP712Sighash
  PASS Random valid init (371ms)
  PASS Random valid refund (231ms)
```

```
  Escrow
    PASS Parse flags (55ms)
    PASS Total deposit calculation
    PASS Random hash

  EscrowStorage
    PASS Commit
    PASS Invalid commit twice
    PASS Commit 2 different
    PASS Commit & finalize success
    PASS Commit & finalize not success
    PASS Invalid commit, finalize, try to re-commit
    PASS Invalid finalize, not committed
    PASS Commit 2 different, finalize 1

  LpVault
    PASS Deposit
    PASS 2 Deposits
    PASS Invalid deposit not enough funds
    PASS Invalid deposit not enough allowance
    PASS Deposit ETH
    PASS Deposit ETH, more than required
    PASS Invalid deposit ETH, too low tx.value
    PASS Invalid deposit ETH, not enough balance
    PASS Withdraw
    PASS Withdraw ETH
    PASS Withdraw all
    PASS Invalid withdraw more than deposited
    PASS Invalid withdraw more than deposited (contract has enough token)
    PASS Transfer out
    PASS Transfer in
    PASS Transfer in all
    PASS Invalid transfer in, more than owned

  ReputationState
    PASS Test updates
    PASS Test overflowing

  ReputationTracker
    PASS Update reputation
    PASS Update reputation multiple types
    PASS Update reputation multiple tokens
    PASS Update reputation multiple claim handlers
    PASS Update reputation multiple tokens & claim handlers
    PASS Update reputation type out of bounds

  Execution
    PASS Random hash

  ExecutionContract(unit)
    PASS Random hash

  ExecutionProxy
    PASS Valid execute
    PASS Valid execute (payable)
    PASS Valid execute (revert)
    PASS Valid execute (contract doesn't exist)
Killed
```